

LYRA: GENERATIVE 3D SCENE RECONSTRUCTION VIA VIDEO DIFFUSION MODEL SELF-DISTILLATION

Anonymous authors
Anonymous submission

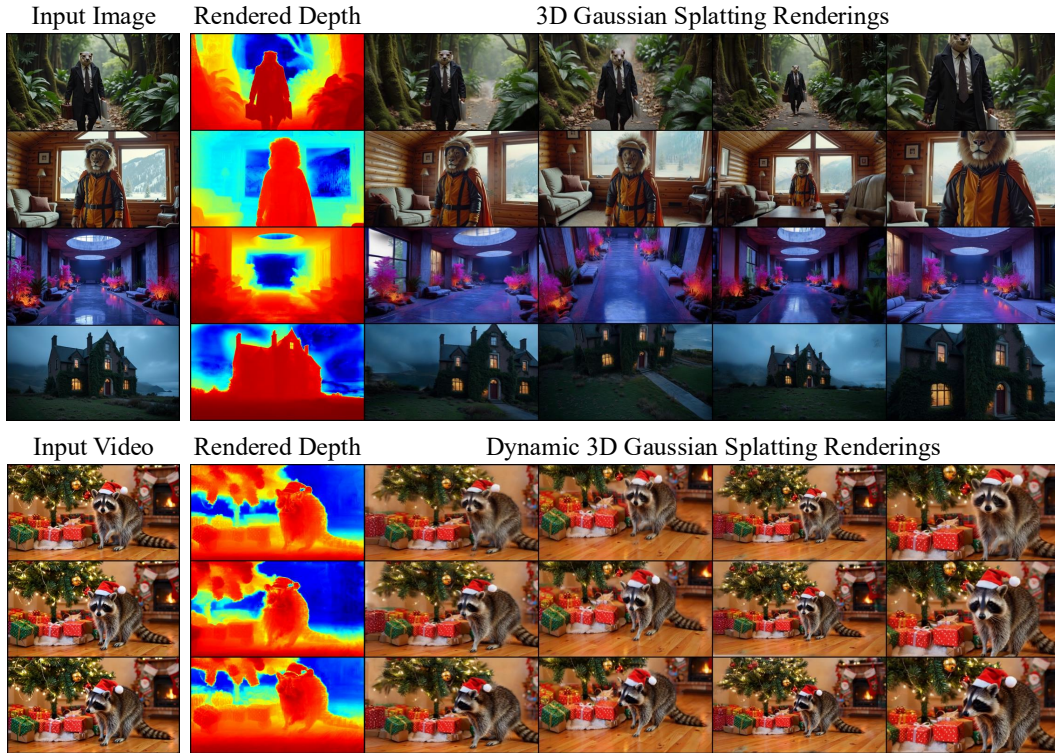


Figure 1: **Feed-Forward 3D and 4D Scene Generation.** From a single image (top), *Lyra* infers a 3D Gaussian Splatting (3DGS) representation in a single feed-forward pass by self-distilling a video diffusion model, without any real-world multi-view data and without per-scene optimization. Given a video input (bottom), *Lyra* infers a dynamic 3DGS that affords interactive control over both time (rows) and viewpoint (columns).

ABSTRACT

The ability to synthesize virtual environments at scale is a foundational capability for applications spanning gaming and the physical-AI domains of robotics, autonomous driving, and industrial AI. Prevailing learning-based 3D reconstruction methods, however, remain fundamentally contingent on captured real-world multi-view data—a resource that is neither uniformly available nor economically scalable. Video diffusion models have recently exhibited a remarkable capacity for visual imagination, yet their intrinsically 2D output precludes the direct geometric use that simulation demands, where an agent must navigate and physically interact with its surroundings. We close this gap with a self-distillation framework that distills the implicit 3D knowledge embedded within video diffusion models into an explicit 3D Gaussian Splatting (3DGS) representation, thereby eliminating the dependence on multi-view training data altogether. Concretely, we augment the standard RGB decoder with a parallel 3DGS decoder supervised directly by the RGB decoder’s output, so that the 3DGS decoder is trained end-to-end from synthetic data generated by the video diffusion model itself. At inference, the model

synthesizes an explicit 3D scene from either a text prompt or a single image and renders it in real time, and the same framework extends naturally to dynamic 3D scene generation from a monocular input video. Across both static and dynamic settings, our framework establishes state-of-the-art 3D scene generation.

1 INTRODUCTION

Synthesizing high-quality 3D environments at scale is a long-standing challenge in computer vision and graphics, and one that underpins applications across film-making, VR/AR, and physical AI with closed-loop simulation. These applications share a common requirement: an explicit 3D representation that simultaneously supports real-time rendering, physical interaction, and geometrically consistent multi-view synthesis.

Recent advances in neural 3D reconstruction (Mildenhall et al., 2020; Kerbl et al., 2023; Moenne-Loccoz et al., 2024) recover precisely such representations from posed images. Their scalability, however, is fundamentally constrained by a reliance on accurate camera poses and high-quality imagery, a difficulty that compounds for dynamic scenes, which typically demand synchronized multi-camera rigs (Li et al., 2022). More fundamentally, reconstruction is confined to observed content and cannot extrapolate beyond the input views. Feed-forward reconstruction models (Zhang et al., 2024e; Ren et al., 2024c) were introduced to relax the pose and per-scene-optimization constraints, yet they inherit a bottleneck of their own: the scarcity of diverse, large-scale 3D training data, which sharply curtails out-of-domain generalization.

Video diffusion models (Cosmos, 2025; Wan, 2025) offer a compelling route past this impasse. Trained on internet-scale video corpora, they attain high fidelity and broad generalization across markedly diverse environments. Because they learn from real-world footage spanning a wide gamut of camera trajectories—from handheld motion to cinematic panning and drone flyovers—these models implicitly encode rich cues about the underlying 3D world without ever observing explicit multi-view supervision. Moreover, and unlike reconstruction approaches (Charatan et al., 2024), generative models can synthesize plausible content beyond what the input frames reveal. The obstacle is that video diffusion models emit only 2D frames and carry no explicit 3D representation, which precludes their direct use in simulation and in downstream tasks that demand geometric consistency, long-term coherence, and physical interaction. Recent work (Wang et al., 2024e; Ren et al., 2025), however, establishes that video models can be endowed with explicit camera control using only modestly sized posed datasets. This capacity to generate *posed* image sequences is pivotal: it converts a video model into a resource that serves simultaneously as the input to and the supervision for 3D reconstruction.

Building on this insight, we unify the hitherto separate paradigms of reconstruction and generation within a single formulation that we term *generative 3D scene reconstruction*. We introduce *Lyra*, a method that generates explicit 3D environments directly from the latent representations of a video diffusion model in a single forward pass. At the core of *Lyra* is a self-distillation framework (Fig. 2) in which a 3D Gaussian Splatting (3DGS) decoder operates directly within the latent space of the video diffusion model. Given a single input image or video, we sample a camera trajectory as conditioning, denoise the resulting video latent, and decode it along two parallel branches: the standard RGB decoder synthesizes a video sequence, while our 3DGS decoder produces an explicit 3D representation. The two branches close a self-distillation loop in which the RGB branch (teacher) supervises the 3DGS branch (student). To broaden viewpoint coverage, we sample multiple camera trajectories, generate the corresponding video latents, and train the 3DGS decoder to fuse information across them while reconciling long-term and multi-view inconsistencies.

This generative reconstruction framework confers three decisive advantages. **(i)** It generates large-scale synthetic environments spanning diverse scenarios directly from video diffusion models, dispensing entirely with real-world multi-view captures. **(ii)** By operating in the latent space of the video model, it aggregates many views efficiently, sidestepping the prohibitive memory overhead that burdens pixel-space feed-forward reconstruction methods. **(iii)** Its explicit 3DGS output enforces geometric consistency and yields representations that are immediately usable in downstream tasks such as physical simulation. As a direct consequence, at inference *Lyra* produces high-quality 3D Gaussian scenes from monocular input that render in real time *without any additional optimization or post-processing*.

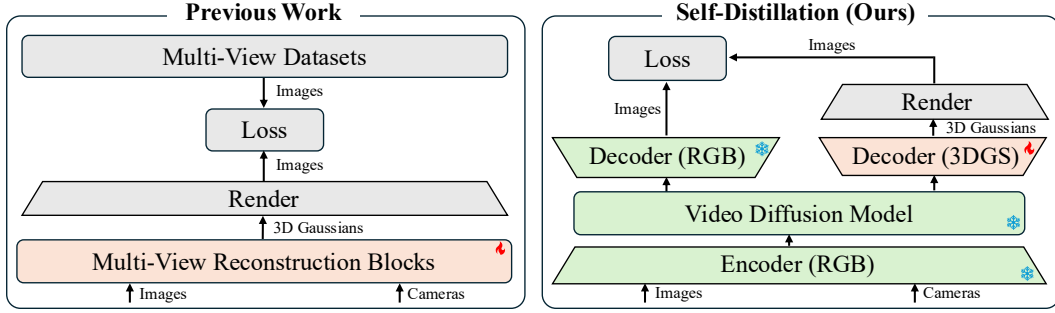


Figure 2: **Self-distillation framework of *Lyra***. Prior work (left) (Szymanowicz et al., 2025b) trains multi-view reconstruction blocks on real-world datasets of limited diversity (Zhou et al., 2018; Ling et al., 2024b). We instead propose a self-distillation framework (right) for generative 3D scene reconstruction, in which the RGB-decoder output (teacher) of a pre-trained camera-controlled video diffusion model supervises the renderings of the 3DGS decoder (student). Because the video model is pre-trained on diverse 2D video, it supplies correspondingly diverse multi-view supervision, and the reliance on captured multi-view data is removed entirely.

We further extend the self-distillation framework to dynamic 4D generation from monocular video. Here the video model (teacher) supplies space-time supervision, while the student learns time-conditioned 3DGS representations that enable novel-view synthesis of dynamic scenes—a capability that follows from the static formulation with only minimal architectural change.

In summary, this work makes the following contributions:

- We introduce a self-distillation framework that trains a 3DGS student decoder using a pre-trained camera-controlled video diffusion model as the teacher, thereby removing the dependence on captured multi-view real-world data entirely.
- We show that the same framework extends to dynamic scenes, enabling feed-forward 4D reconstruction from a monocular video—a setting left unexplored by prior feed-forward methods.
- We demonstrate that the resulting model generalizes across diverse scenes, establishing state-of-the-art results in both single-image 3D scene generation and single-video 4D scene generation.

2 RELATED WORK

Multi-view image generation. Early work on multi-view image generation (Watson et al., 2023; Liu et al., 2023; Shi et al., 2024; Wang & Shi, 2023) concentrated on object-centric scenes devoid of background, and subsequent efforts (Sargent et al., 2024; Tang et al., 2023b; Schneider et al., 2025) scaled multi-view image generators up to full scenes. CAT3D (Gao et al., 2024b) repurposes a pre-trained image diffusion model to generate multi-view images from any generated or real input image, after which a second stage reconstructs the generated views into an explicit representation such as NeRF (Mildenhall et al., 2020) or 3DGS (Kerbl et al., 2023). This paradigm has since been extended to dynamic 3D scenes rendered from multiple viewpoints via space-time diffusion models (Watson et al., 2024; Wu et al., 2025b; Kuang et al., 2024; Wang et al., 2025a). Impressive as these results are, grounding the generations in an explicit 3D representation hinges on an expensive per-scene optimization stage whose cost is not amortized across scenes. Rather than synthesizing multi-view images and subsequently reconstructing them by optimization, we generate a 3D scene *directly* from text or a single image, obviating the per-scene optimization stage entirely.

Camera-conditioned video generation. Fine-tuning video diffusion models for explicit 3D camera control has advanced rapidly. MotionCtrl (Wang et al., 2024e) pioneered this direction by conditioning pre-trained video models on camera poses. Follow-up work (He et al., 2024a; Xu et al., 2024c; Bahmani et al., 2025a;b) encodes cameras as Plücker coordinates for pixel-wise conditioning, while a complementary line (Hu et al., 2024; Hou et al., 2024; Zhang et al., 2024b) pursues training-free camera control of video diffusion models. More recently, SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a) achieved compelling camera-controlled video synthesis from large-scale synthetic training data. Compelling as they are, the outputs of these methods remain ungrounded in 3D. Our work is orthogonal and complementary: we address feed-forward 3D generation *from* camera-controlled video models, so that novel viewpoints can be rendered *after* generation without further optimization. Concretely, we build upon GEN3C (Ren et al., 2025), a recent camera-controlled

video diffusion model, and employ it as the teacher within our self-distillation framework for 3D reconstruction.

Feed-forward 3D models. Early feed-forward 3D reconstruction targeted object-centric scenes without background, predicting a NeRF (Hong et al., 2024; Li et al., 2024b) or 3DGS (Tang et al., 2024a; Szymanowicz et al., 2024) directly from text or an image. More recent methods (Charatan et al., 2024; Zhang et al., 2024e; Szymanowicz et al., 2025a; Ren et al., 2024c) address scene-level reconstruction, typically regressing per-pixel 3D Gaussians from one or more input images. Most such scene-level methods, however, remain tethered to their training distribution—e.g., RealEstate10K (Zhou et al., 2018)—and generalize poorly to generated scenes. A further line of work pushes feed-forward reconstruction toward scene *generation*: Bolt3D (Szymanowicz et al., 2025b) trains a pointmap (Wang et al., 2024b) autoencoder to jointly generate pointmaps and multi-view images for feed-forward reconstruction, and, most closely related to us, Wonderland (Liang et al., 2025a) uses a camera-controlled video model to produce 3D Gaussians with a feed-forward network. In contrast, we treat a camera-controlled video model as a teacher within a self-distillation framework, and thereby train our student 3D model *without any real-world multi-view data*. Crucially, we further show that this framework admits a simple extension to feed-forward 4D scene generation, a task that has remained unexplored to date.

3 SELF-DISTILLATION USING VIDEO DIFFUSION MODELS

The central idea of our approach is to distill the implicit 3D knowledge embedded in video diffusion models into an explicit 3DGS decoder that produces high-quality 3D representations. We realize this idea through a teacher–student framework in which the video diffusion model (teacher) generates RGB videos that supervise a 3DGS decoder (student) operating within the same latent space, as illustrated in Fig. 2. This section first details the video diffusion model backbone (Sec. 3.1) and its role as the teacher in self-distillation for the 3DGS decoder (Sec. 3.2). We then motivate and specify the design of the 3DGS decoder in Sec. 4, and derive the minimal modifications that adapt the pipeline to dynamic 3D scenes in Sec. 5.

3.1 BACKGROUND: CAMERA-CONTROLLED AND 3D-CONSISTENT VIDEO DIFFUSION

Because our 3DGS decoder is supervised exclusively by the camera-controlled video diffusion model, the 3D consistency of that model is a strict prerequisite for our method. We therefore build on GEN3C (Ren et al., 2025), a recent camera-conditioned video diffusion model, and summarize below the key design choices on which our formulation depends.

Spatiotemporal 3D cache. To sharpen both video consistency and camera-control precision, GEN3C constructs a spatiotemporal 3D cache $\{\mathbf{P}^{t,v}\}$ from the input image(s) or videos, where each $\mathbf{P}^{t,v}$ is a colored point cloud obtained by unprojecting the estimated depth (Wang et al., 2024a) of an RGB image captured from camera viewpoint v at time t . The cache is organized as an $L \times V$ array, in which L denotes the temporal length (number of frames) and V the number of camera views.

Rendering and structured guidance. To exploit this cache, GEN3C renders each point cloud from an arbitrary target camera pose, yielding RGB images $\mathbf{I}^{t,v}$ and disocclusion masks $\mathbf{M}^{t,v}$ through $(\mathbf{I}^{t,v}, \mathbf{M}^{t,v}) = \mathcal{R}(\mathbf{P}^{t,v}, \mathbf{C}^t)$, where the rendering function $\mathcal{R}$ projects the 3D point cloud $\mathbf{P}^{t,v}$ onto the 2D image plane according to the camera pose $\mathbf{C}^t$ at time t . Given a sequence of camera poses $\mathbf{C} = (\mathbf{C}^1, \dots, \mathbf{C}^L)$, rendering every cache element produces V videos of length L , which stack into per-view image sequences $\mathbf{I}^v \in \mathbb{R}^{L \times 3 \times H \times W}$ and mask sequences $\mathbf{M}^v \in \mathbb{R}^{L \times 1 \times H \times W}$. The disocclusion masks delineate precisely the regions that the video diffusion model must inpaint; together, the renderings and masks constitute the structured visual guidance that conditions the subsequent video generation.

Video variational autoencoder (VAE). Diffusion-based video generators conventionally rely on a video variational autoencoder (VAE) (Kingma & Welling, 2013; Rombach et al., 2022) to compress videos into a lower-dimensional latent space, rendering both training and inference tractable. Given an RGB video $\mathbf{I} \in \mathbb{R}^{L \times 3 \times H \times W}$, a pre-trained VAE encoder $\mathcal{E}$ maps it into this latent space as $\mathbf{z} = \mathcal{E}(\mathbf{I}) \in \mathbb{R}^{L' \times C \times h \times w}$, within which the diffusion model is trained and sampled. The generated video $\hat{\mathbf{I}} = \mathcal{D}_{rgb}(\mathbf{z})$ is then recovered by a pre-trained VAE decoder $\mathcal{D}_{rgb}$. We adopt the pre-trained GEN3C (Ren et al., 2025; Cosmos, 2025) model, for which the latent channel dimension is $C = 16$, the temporal dimension is $L' = (L - 1)/\tau + 1$, and the spatial dimensions are $h = H/\sigma$ and

$w = W/\sigma$, with temporal compression factor $\tau = 8$ and spatial compression factor $\sigma = 8$. This compression is precisely what renders the many-view regime tractable for our decoder, as we exploit in Sec. 4.

3.2 SELF-DISTILLATION

We train the 3DGS decoder $\mathcal{D}_s$ as the student in a teacher–student paradigm whose teacher is a camera-controlled video diffusion model $\mathcal{V}$. To secure diverse supervision, we curate a large-scale collection of text prompts with large language models (OpenAI, 2025; Bai et al., 2023), synthesize images I with an image diffusion model (Black Forest Labs, 2024), and expand each image into multi-view sequences using GEN3C (Ren et al., 2025). This pipeline furnishes supervision whose diversity is inherited directly from the video model’s internet-scale pre-training rather than from any curated multi-view corpus.

Teacher–student setup. Given an input image I and a sampled camera trajectory $\{\mathbf{C}^t\}_{t=1}^L$, the video diffusion model $\mathcal{V}$ produces a denoised video latent $\mathbf{z} = \mathcal{V}(I, \{\mathbf{C}^t\}_{t=1}^L)$. This latent is decoded along two parallel branches: the pre-trained RGB decoder $\mathcal{D}_{rgb}$ yields video frames $\mathbf{I}_{\mathcal{D}_{rgb}} = \mathcal{D}_{rgb}(\mathbf{z})$, while the 3DGS decoder $\mathcal{D}_s$ emits explicit 3D Gaussians $\mathbf{G}$. The views rendered from the Gaussians (*student*), $\mathbf{I}_{\mathcal{D}_s} = \text{Render}(\mathbf{G}, \{\mathbf{C}^t\}_{t=1}^L)$, are supervised to match the RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}$ (*teacher*), thereby closing the self-distillation loop: the teacher supplies photometric targets, and the student is compelled to explain them with a single, view-consistent explicit 3D representation.

Multi-trajectory supervision. To enlarge the viewpoint coverage afforded by a single input image, we sample $V = 6$ camera trajectories per image, as shown in Fig. 3. For each trajectory v we construct a spatiotemporal cache $\{\mathbf{P}^{t,v}\}_{t=1}^L$ together with $L = 121$ camera poses $\{\mathbf{C}^{t,v}\}_{t=1}^L$. Passing these caches through the video diffusion model $\mathcal{V}$ yields latents $\mathbf{z}^v$, which the teacher decodes into RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}^v = \mathcal{D}_{rgb}(\mathbf{z}^v)$. The 3DGS decoder $\mathcal{D}_s$ is then trained to fuse the multiple $\mathbf{z}^v$ into a single coherent set of Gaussians $\mathbf{G}$ while inpainting disoccluded regions, reconciling the residual inconsistencies that arise across independently sampled trajectories. This design trains $\mathcal{D}_s$ entirely from synthetic supervision provided by $\mathcal{V}$, and its two properties are complementary rather than incidental: operating in latent space permits efficient aggregation across many trajectories, whereas the explicit Gaussian output $\mathbf{G}$ enforces geometric consistency and directly supports downstream physical simulation and real-time rendering.

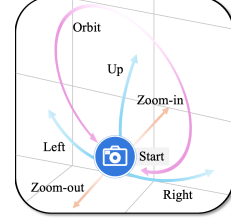


Figure 3: The six sampled camera trajectories per input that jointly maximize viewpoint coverage for multi-trajectory supervision.

4 FEED-FORWARD RECONSTRUCTION FROM MULTI-VIEW VIDEO LATENTS

Our feed-forward 3DGS decoder $\mathcal{D}_s$ transforms the multi-view latents synthesized by the video model $\mathcal{V}$ into an explicit 3D representation renderable from arbitrary viewpoints. We adopt 3D Gaussian Splatting (3DGS) (Kerbl et al., 2023) for its explicit parameterization and real-time rendering, and detail the decoder architecture in Fig. 4 (right).

4.1 3DGS DECODER

Scaling Latent-Based 3D Reconstruction. Prior feed-forward reconstruction frameworks generate one 3D Gaussian per input pixel and are therefore bounded by the number and resolution of views they can ingest: GS-LRM (Zhang et al., 2024e) operates on 2–4 images at 512×512 , and AnySplat (Jiang et al., 2025a) trains on 24 images at 448×448 . By contrast, our video model synthesizes $V \times L = 6 \times 121 = 726$ input views at 704×1280 resolution, a regime that far exceeds the capacity of existing methods and overruns available GPU memory. The bottleneck is the attention over visual tokens, whose memory and compute cost scale with the pixel count. We eliminate it by declining to scale in pixel space at all, operating instead directly in the compressed video latent space produced by our camera-controlled video diffusion model—the same $8 \times$ spatiotemporal compression introduced in Sec. 3.1. We denote the resulting multi-view video latents as $\mathbf{Z} \in \mathbb{R}^{V \times L' \times C \times h \times w}$.

Architecture. The 3DGS decoder $\mathcal{D}_s$ maps its inputs into the hidden dimension of the main reconstruction blocks and predicts per-pixel 3D Gaussian features $\mathbf{G} \in \mathbb{R}^{V \times L \times H \times W \times 14}$. Its inputs

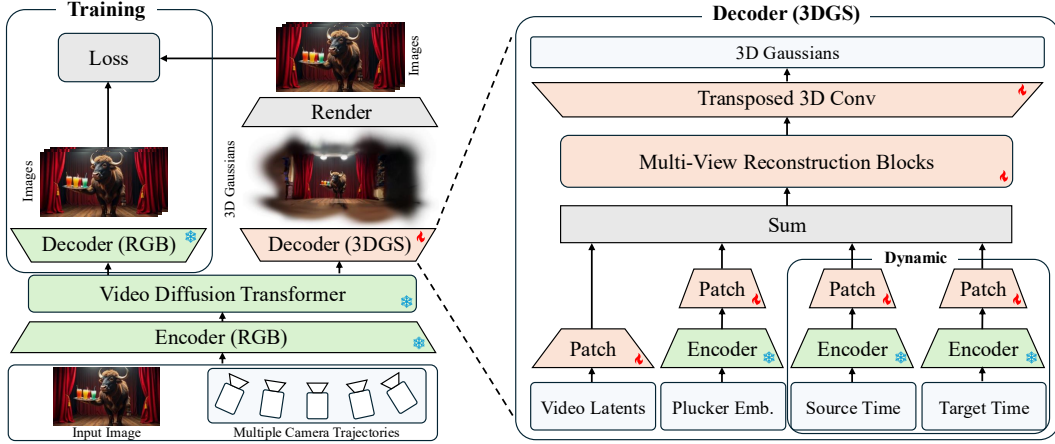


Figure 4: **3D Generative Reconstruction Framework.** Our pipeline builds upon a camera-controlled video diffusion model (Ren et al., 2025) pre-trained on large-scale data. We train a 3D Gaussian Splatting (3DGS) decoder by aligning the 2D renderings of the generated 3DGS scenes with the RGB-decoded outputs of the pre-trained video model, updating only the 3DGS decoder while keeping the pre-trained autoencoder and diffusion model frozen. At inference, the 3DGS decoder is used directly and the RGB decoder is no longer required. Time conditioning within the 3DGS decoder further extends the approach seamlessly from static to dynamic 3D scene generation, with no change to the underlying architecture.

are the multi-view video latents $\mathbf{Z}$ and the encoded Plücker embeddings $\mathbf{E}$, both of latent dimension $\mathbf{Z}, \mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. Each input is patchified to the block hidden dimension by a single spatial 2×2 patchification layer (Dosovitskiy et al., 2021); crucially, no additional visual encoder is required, since the video latents already constitute a compact token representation. The two patchified streams are summed and propagated through the reconstruction blocks. Following the design of Long-LRM (Ziwen et al., 2024), each block comprises one Transformer (Vaswani et al., 2017) layer followed by seven Mamba-2 (Dao & Gu, 2024) layers, and we stack the block twice to obtain 16 layers with a 512-dimensional hidden state—an interleaving that couples the global receptive field of attention with the near-linear sequence scaling of Mamba-2. A final transposed 3D convolution maps the hidden representation to the 14 Gaussian channels: 3D position (x, y, z) , scale (s_x, s_y, s_z) , rotation quaternion (q_w, q_x, q_y, q_z) , opacity α , and RGB (r, g, b) . The static decoder is thus expressed compactly as $\mathbf{G} = \mathcal{D}_s(\mathbf{Z}, \mathbf{E})$.

Plücker embeddings. To supply the decoder with explicit camera geometry, we first compute raw Plücker embeddings from the camera poses $\{\mathbf{C}^{t,v}\}$ and intrinsics as $\mathbf{E}^{\text{raw}} \in \mathbb{R}^{V \times L \times 6 \times H \times W}$, and encode them by reusing the RGB encoder $\mathcal{E}$ of the pre-trained VAE, so that the camera representation resides in the same latent space as $\mathbf{Z}$. Since $\mathcal{E}$ accepts three-channel input, we encode the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins separately, then concatenate the two latents along the channel dimension to obtain $\mathbf{E}^{\text{enc}} \in \mathbb{R}^{V \times L' \times 2C \times h \times w}$. A lightweight MLP finally maps $\mathbf{E}^{\text{enc}}$ to $\mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$, halving the channel dimension so that it aligns with the video latent $\mathbf{Z}$.

4.2 LOSS FUNCTION

Image-based supervision. We supervise the 3DGS decoder with an image-based reconstruction objective that decomposes into a Mean Squared Error (MSE) term $\mathcal{L}_{mse}$, which enforces pixel-wise photometric fidelity, and a perceptual LPIPS term $\mathcal{L}_{lpips}$ (Zhang et al., 2018), computed with VGG (Simonyan & Zisserman, 2014) as the feature extractor to preserve perceptual structure.

Depth supervision. A decoder trained under RGB supervision alone tends to collapse geometry into flattened surfaces. To counteract this failure mode, we additionally supervise the rendered depth maps against the temporally consistent video depth estimated by the off-the-shelf system ViPE (Huang et al., 2025), using the scale-invariant depth loss $\mathcal{L}_{depth}$ of Long-LRM (Ziwen et al., 2024).

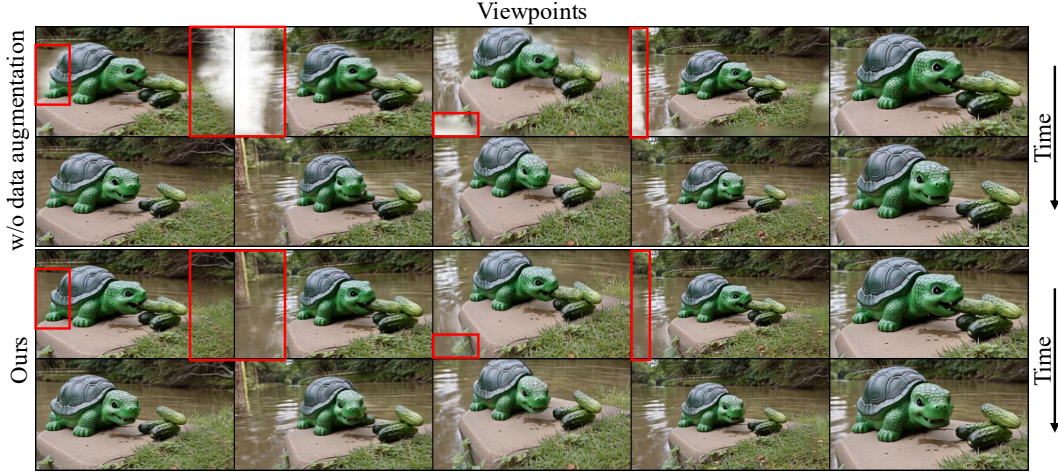


Figure 5: **Dynamic data augmentation.** Naively supervising the time-aware 3DGS decoder induces artifacts in the generated 3D Gaussians: at early timesteps, regions uncovered by the supervision signal collapse to low opacity under extreme poses. Augmenting the supervision with a motion-reversed video guarantees that every timestep is observed with full spatial coverage, thereby eliminating these low-opacity artifacts at early timesteps.

Opacity-based pruning. Following Long-LRM (Ziwen et al., 2024), we impose an L_1 regularizer on opacity, $\mathcal{L}_{opacity}$, and prune the Gaussians occupying the lowest 80% of opacity, which yields a substantially more compact representation without loss of fidelity.

Total loss. The total objective is the weighted sum of the individual terms, with weighting factors λ_i ,

$$\mathcal{L} = \lambda_{mse} \mathcal{L}_{mse} + \lambda_{lpips} \mathcal{L}_{lpips} + \lambda_{depth} \mathcal{L}_{depth} + \lambda_{opacity} \mathcal{L}_{opacity} \quad (1)$$

where we set $\lambda_{mse} = 1.0$, $\lambda_{lpips} = 0.5$, $\lambda_{depth} = 0.05$, and $\lambda_{opacity} = 0.1$ throughout.

5 EXTENSION TO DYNAMIC 3D SCENES

A central strength of our framework is that it extends to dynamic 3D scenes with only minimal changes, which we detail below.

Self-distillation for dynamic scenes. The dynamic setup mirrors its static counterpart closely. In place of a single image, the video model now ingests a single video of length L together with its camera poses and generates multi-view video latents that capture the same underlying motion. For video inputs we follow a protocol analogous to the static case: we sample diverse text prompts with large language models and generate videos with video diffusion models (Cosmos, 2025; Wan, 2025), and subsequently annotate each generated video with camera poses and depth maps via ViPE (Huang et al., 2025).

Dynamic 3DGS decoder architecture. Adopting the bullet-time formulation of Liang et al. (2025b), we design the dynamic 3DGS decoder $\mathcal{D}_s$ to predict time-dependent 3D Gaussians for every video frame. The decoder $\mathcal{D}_s$ retains the architecture of $\mathcal{D}_s$ and augments its input with encoded source and target time embeddings $\mathbf{T}^{src}, \mathbf{T}^{tgt} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The raw source times are assigned per input frame as $\mathbf{T}^{src, raw} \in \mathbb{R}^{V \times L \times 1 \times H \times W}$, and the raw target time as $\mathbf{T}^{tgt, raw} \in \mathbb{R}^{V \times 1 \times 1 \times H \times W}$. Both are spatially replicated and combined with a 2-dimensional sinusoidal embedding to expand the channel dimension to 3, then encoded by the RGB encoder $\mathcal{E}$ into latents $\mathbf{T}^{src}, \mathbf{T}^{tgt} \in \mathbb{R}^{V \times L' \times C \times h \times w}$, with the target time replicated along the latent temporal dimension to match L' . At training time we sample a target timestep at random, decode the corresponding 3D Gaussians, and supervise them across all trajectories produced by dynamic data augmentation. We fine-tune $\mathcal{D}_d$ from the pre-trained $\mathcal{D}_s$, initializing the patchification layers of $\mathbf{T}^{src}$ and $\mathbf{T}^{tgt}$ to zero so that training begins from the exact static behavior and departs from it only as the time conditioning becomes informative; $\mathbf{T}^{src}$ and $\mathbf{T}^{tgt}$ are then added directly to the sum of $\mathbf{Z}$ and $\mathbf{E}$. The dynamic decoder is thus defined as $\mathbf{G} = \mathcal{D}_d(\mathbf{Z}, \mathbf{E}, \mathbf{T}^{src}, \mathbf{T}^{tgt})$.

Table 1: **State-of-the-art comparisons.** Single-image-to-3D generation on RealEstate10K, DL3DV, and Tanks-and-Temples, evaluated with PSNR, SSIM, and LPIPS. Our method attains the best value on every metric of every dataset, uniformly surpassing all prior methods; the best value in each column is shown in **bold**.

Method	RealEstate10K			DL3DV			Tanks-and-Temples		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
ZeroNVS	13.01	0.378	0.448	13.35	0.339	0.465	12.94	0.325	0.470
ViewCrafter	16.84	0.514	0.341	15.53	0.525	0.352	14.93	0.483	0.384
Wonderland	17.15	0.550	0.292	16.64	0.574	0.325	15.90	0.510	0.344
Bolt3D	21.54	0.747	0.234	-	-	-	-	-	-
Ours	21.79	0.752	0.219	20.09	0.583	0.313	19.24	0.570	0.336

Dynamic data augmentation. In static scenes, where time plays no role, frames from every timestep are jointly valid supervision for the 3DGS renderings. In dynamic scenes, by contrast, the 3DGS evolves over time, so only frames from the matching timestamp constitute valid supervision. This constraint admits a degenerate solution in which the 3DGS at a given timestep simply disregards information from frames at other timesteps (see Fig. 5).

We resolve this failure mode with a dynamic data augmentation strategy that balances supervision across timesteps. Early frames are naturally tied to viewpoints near the input image, whereas later frames correspond to more distant viewpoints; to redress this imbalance, we furnish each timestep with paired supervision views, one near and one far. Concretely, we reverse the input video in frame order and feed it to the video model $\mathcal{V}$, obtaining six additional multi-view sequences. Reversing these sequences back to the original motion order yields six inward-progressing trajectories (from distant viewpoints toward the input) which, combined with the original six outward trajectories, provide 12 supervision views per timestep. This augmentation is confined to training: the reversed trajectories serve purely as supplementary supervision that forestalls collapse from pruning artifacts, and are not required at inference.

6 EXPERIMENTS

6.1 EXPERIMENTAL SETUP

Datasets. We train our model without recourse to any existing multi-view dataset; instead, we construct our own, the *Lyra* dataset, supervising the 3DGS decoder entirely from our video model. The 3D reconstruction setup comprises 59,031 images and the 4D setup 7,378 videos, all originating from diverse text prompts that span indoor and outdoor environments, humans, animals, and both realistic and imaginative content. Synthesizing six camera trajectories per image (3D) or video (4D) yields 354,186 videos for the 3D setup and 44,268 videos for the 4D setup, demonstrating that the self-distillation pipeline scales the supervision set far beyond what multi-view capture would permit.

Baselines. We benchmark our method against the state-of-the-art approaches for generative single-image-to-3D reconstruction: ZeroNVS (Sargent et al., 2024), ViewCrafter (Yu et al., 2024b), Wonderland (Liang et al., 2025a), and Bolt3D (Szymanowicz et al., 2025b). Because *no source code is available* for these methods, we cannot re-evaluate them on our out-of-distribution set and therefore compare against the quantitative results reported in the respective papers.

Evaluation. Consistent with prior work, we evaluate single-image-to-3D on RealEstate10K (Zhou et al., 2018), DL3DV (Ling et al., 2024b), and Tanks-and-Temples (Knapitsch et al., 2017), adopting the evaluation protocol of Wonderland (Liang et al., 2025a) and Bolt3D (Szymanowicz et al., 2025b) and reporting the standard reconstruction metrics PSNR, SSIM, and LPIPS. This protocol places our method and prior work on identical footing across three benchmarks of contrasting scene statistics.

6.2 MAIN RESULTS

Quantitative results. Tab. 1 reports the single-image-to-3D evaluation. Our method surpasses all prior work on every benchmark and across every metric, attaining higher PSNR and SSIM and lower LPIPS on RealEstate10K, DL3DV, and Tanks-and-Temples alike; the advantage holds uniformly and is most pronounced on DL3DV and Tanks-and-Temples, the two benchmarks that depart furthest



Figure 6: **Image-to-3DGS Generation.** Five novel views rendered from our generated 3DGS scenes, spanning diverse indoor, outdoor, realistic, and imaginative content and remaining faithful to the input across wide viewpoint changes.



Figure 7: **Ablations.** Each ablation is visualized by rendering the same extreme novel viewpoint after image-to-3DGS generation, making the degradation from removing each component directly apparent; depth visualizations and further ablations appear in Appendix C.4.

from the indoor scene statistics of RealEstate10K. Because the reconstruction is distilled from the video model, this consistency also charts a clear path to further gains: stronger video generative backbones will translate directly into improved reconstructions, without any change to the decoder or its training. Additional quantitative comparisons on the *Lyra* dataset are provided in Appendix C.

Qualitative results. Fig. 1 and Fig. 6 present novel-view renderings from our generated 3DGS scenes; video results will be released upon acceptance. Our method synthesizes high-quality novel-view content in previously unobserved regions of both 3D and 4D scenes while remaining faithful to the input image or video, corroborating the quantitative gains above. Qualitative comparisons on the *Lyra* dataset appear in Appendix C.

6.3 ABLATIONS

Tab. 2 substantiates our design choices by ablating each key component on out-of-distribution diverse prompts, evaluating the 3DGS renderings against the videos generated by our camera-controlled video diffusion model. Every ablation degrades performance relative to the full model, confirming that each component is necessary rather than incidental to the reported quality.

Real data only. Replacing self-distillation with training on only the real multi-view datasets RealEstate10K (Zhou et al., 2018) and DL3DV (Ling et al., 2024b)—the datasets favored in prior work—sharply degrades generalization to out-of-distribution scenes, from 24.77 to 19.08 PSNR. This 5.69 dB gap is the single largest among all data and loss ablations, and directly quantifies the value of distilling supervision from the video model rather than from captured multi-view data.

Self-distillation + real data. Notably, augmenting self-distillation with real data yields no improvement over self-distillation alone (24.74 vs. 24.77 PSNR). This confirms that the synthetic supervision from self-distillation is already sufficiently diverse and consistent to train the re-

Table 2: **Ablation study on the *Lyra* dataset.** Removing any single component—self-distillation data, a loss term, or an architectural element—degrades performance relative to the full model (top row), confirming that each contributes to the reported quality; the largest drops trace to the loss of self-distillation and of learned multi-view fusion.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Ours	24.77	0.837	0.224

construction model on its own, and that real multi-view data is not merely unnecessary but adds nothing once self-distillation is in place.

No depth loss. The depth loss prevents flattened geometry and even marginally improves the image-based metrics (24.77 vs. 24.31 PSNR), so geometric fidelity is obtained at no cost to appearance; corresponding depth maps are visualized in Appendix C.4.

No opacity pruning. Retaining all Gaussians rather than pruning by opacity is both less compact and slightly worse in visual quality: learning to prune the majority of the Gaussians yields a more compact representation while simultaneously improving quality. Moreover, at $H = 704$ and $W = 1280$, rendering takes 18ms with pruning versus 30ms without—a $1.67\times$ speedup—so pruning improves quality, compactness, and rendering speed at once.

No LPIPS loss. The LPIPS loss confers robustness to input inconsistencies and better preserves high-frequency detail; its removal degrades LPIPS markedly, from 0.224 to 0.370.

No multi-view fusion. A naive alternative generates 3D Gaussians independently per camera trajectory and fuses the resulting point clouds post hoc. Learning the multi-view fusion instead within our reconstruction blocks, where each token attends to all others, is decisively better, raising PSNR from 17.73 to 24.77—a 7.04 dB improvement that identifies learned latent-space fusion as the most consequential architectural choice.

No Mamba-2. Replacing our joint Transformer/Mamba-2 blocks with Transformer-only blocks lowers quality slightly while incurring a substantial efficiency penalty: a single forward pass with $V = 6$, $L = 121$, $H = 704$, and $W = 1280$ takes 3213ms with the joint blocks versus 20922ms with Transformer-only blocks, a $6.5\times$ speedup at no quality cost.

No latent-based 3DGS. Pixel-space methods such as BTimer (Liang et al., 2025b) are confined to 12 input frames, whereas our latent-space formulation accommodates up to 726. Operating the 3DGS decoder in pixel space rather than latent space consequently runs out of memory, establishing that the latent formulation is precisely what makes this input scale feasible.

7 CONCLUSION

We have presented *Lyra*, a unified 3D and 4D generation framework driven by a single image or video. Rather than collecting multi-view datasets, we cast camera-controlled video diffusion models as teachers within a self-distillation framework that trains a student 3DGS decoder operating directly in the video model’s latent space and reconstructing 3D Gaussians without any post-processing or optimization. This design broadens generalization and coverage across diverse scenes, and its dynamic extension establishes feed-forward 4D generation from monocular video as a practical capability, achieving state-of-the-art results in both settings. The scale and consistency of our generated scenes are presently bounded only by the capacity of the underlying camera-controlled video diffusion model; consequently, advances in that backbone will translate directly into larger-scale, more consistent scene synthesis. Two further directions are especially promising: adapting auto-regressive techniques (Chen et al., 2024a) to our framework for large-scale generation, and modeling motion and tracking within the reconstruction network, as in concurrent work (Lin et al., 2025), to further sharpen visual motion quality.

8 ETHICS STATEMENT

As with all generative AI technologies, our method carries a risk of misuse, notably the production of misleading or synthetic 3D/4D content. That risk notwithstanding, the central aim of this research is to advance simulation, robotics, and embodied AI by furnishing scalable and controllable tools for

data generation. We accordingly anticipate that its most significant contributions will arise through the following capabilities and insights.

- **Realistic simulation for embodied agents:** We provide controllable and physically consistent camera motion within generated 3D and 4D environments, thereby supporting the training and evaluation of agents that must perceive, navigate, and interact with complex scenes.
- **Improved scalability for data engines:** By removing the need for multi-view capture and expensive per-scene optimization, our method produces high-fidelity, customizable scenes at scale, a capability particularly well suited to research in robotics, reinforcement learning, and closed-loop simulation.
- **Better technical understanding:** As an academic study, this work contributes a foundational case study of how camera control interacts with spatiotemporal generative models, of direct benefit to research in computer vision, graphics, and embodied AI.

While we caution against deceptive or wrongful use of this technology—where realistic generative outputs could yield misleading or unverifiable media—we hold that the adoption of appropriate safeguards, such as provenance tracking, dataset documentation, and sound evaluation practices, will ensure that generative models advance science and society in positive ways.

9 REPRODUCIBILITY STATEMENT

To enable full reproduction of our results and to accelerate future research in this area, we will release our training and inference code, model weights, and data for both 3D and 4D generation upon acceptance.

REFERENCES

- Sherwin Bahmani, Jeong Joon Park, Despoina Paschalidou, Hao Tang, Gordon Wetzstein, Leonidas Guibas, Luc Van Gool, and Radu Timofte. 3d-aware video generation. In *TMLR*, 2023a.
- Sherwin Bahmani, Jeong Joon Park, Despoina Paschalidou, Xinguang Yan, Gordon Wetzstein, Leonidas Guibas, and Andrea Tagliasacchi. CC3D: Layout-conditioned generation of compositional 3D scenes. In *Proc. ICCV*, 2023b.
- Sherwin Bahmani, Xian Liu, Wang Yifan, Ivan Skorokhodov, Victor Rong, Ziwei Liu, Xihui Liu, Jeong Joon Park, Sergey Tulyakov, Gordon Wetzstein, Andrea Tagliasacchi, and David B. Lindell. Tc4d: Trajectory-conditioned text-to-4d generation. In *Proc. ECCV*, 2024a.
- Sherwin Bahmani, Ivan Skorokhodov, Victor Rong, Gordon Wetzstein, Leonidas Guibas, Peter Wonka, Sergey Tulyakov, Jeong Joon Park, Andrea Tagliasacchi, and David B. Lindell. 4d-fy: Text-to-4d generation using hybrid score distillation sampling. In *Proc. CVPR*, 2024b.
- Sherwin Bahmani, Ivan Skorokhodov, Guocheng Qian, Aliaksandr Siarohin, Willi Menapace, Andrea Tagliasacchi, David B Lindell, and Sergey Tulyakov. Ac3d: Analyzing and improving 3d camera control in video diffusion transformers. *Proc. CVPR*, 2025a.
- Sherwin Bahmani, Ivan Skorokhodov, Aliaksandr Siarohin, Willi Menapace, Guocheng Qian, Michael Vasilkovsky, Hsin-Ying Lee, Chaoyang Wang, Jiaxu Zou, Andrea Tagliasacchi, et al. Vd3d: Taming large video diffusion transformers for 3d camera control. *Proc. ICLR*, 2025b.
- Jianhong Bai, Menghan Xia, Xiao Fu, Xintao Wang, Lianrui Mu, Jinwen Cao, Zuozhu Liu, Haoji Hu, Xiang Bai, Pengfei Wan, et al. Recammaster: Camera-controlled generative rendering from a single video. *arXiv preprint arXiv:2503.11647*, 2025a.
- Jianhong Bai, Menghan Xia, Xintao Wang, Ziyang Yuan, Xiao Fu, Zuozhu Liu, Haoji Hu, Pengfei Wan, and Di Zhang. Syncammaster: Synchronizing multi-camera video generation from diverse viewpoints. *Proc. ICLR*, 2025b.

-
- 594 Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou,
595 and Jingren Zhou. Qwen-vl: A versatile vision-language model for understanding, localization,
596 text reading, and beyond. *arXiv preprint arXiv:2308.12966*, 2023.
- 597
- 598 Black Forest Labs. Flux. <https://github.com/black-forest-labs/flux>, 2024.
- 599
- 600 Andreas Blattmann, Tim Dockhorn, Sumith Kulal, Daniel Mendelevitch, Maciej Kilian, Dominik
601 Lorenz, Yam Levi, Zion English, Vikram Voleti, Adam Letts, et al. Stable video diffusion: Scaling
602 latent video diffusion models to large datasets. *arXiv preprint arXiv:2311.15127*, 2023.
- 603
- 604 Chenjie Cao, Jingkai Zhou, Shikai Li, Jingyun Liang, Chaohui Yu, Fan Wang, Xiangyang Xue, and
605 Yanwei Fu. Uni3c: Unifying precisely 3d-enhanced camera and human motion controls for video
606 generation. *arXiv preprint arXiv:2504.14899*, 2025.
- 607
- 608 Yukang Cao, Liang Pan, Kai Han, Kwan-Yee K Wong, and Ziwei Liu. Avatargo: Zero-shot 4d
609 human-object interaction generation and animation. *arXiv preprint arXiv:2410.07164*, 2024.
- 610
- 611 Zenghao Chai, Chen Tang, Yongkang Wong, and Mohan Kankanhalli. Star: Skeleton-aware text-
612 based 4d avatar generation with in-network motion retargeting. *arXiv preprint arXiv:2406.04629*,
613 2024.
- 614
- 615 Eric R Chan, Connor Z Lin, Matthew A Chan, Koki Nagano, Boxiao Pan, Shalini De Mello, Orazio
616 Gallo, Leonidas J Guibas, Jonathan Tremblay, Sameh Khamis, et al. Efficient geometry-aware 3D
617 generative adversarial networks. In *Proc. CVPR*, 2022.
- 618
- 619 Eric R Chan, Koki Nagano, Matthew A Chan, Alexander W Bergman, Jeong Joon Park, Axel Levy,
620 Miika Aittala, Shalini De Mello, Tero Karras, and Gordon Wetzstein. Generative novel view
621 synthesis with 3d-aware diffusion models. In *Proc. ICCV*, 2023.
- 622
- 623 David Charatan, Sizhe Lester Li, Andrea Tagliasacchi, and Vincent Sitzmann. pixelsplat: 3d gaussian
624 splats from image pairs for scalable generalizable 3d reconstruction. In *Proc. CVPR*, 2024.
- 625
- 626 Boyuan Chen, Diego Martí Monsó, Yilun Du, Max Simchowitz, Russ Tedrake, and Vincent Sitzmann.
627 Diffusion forcing: Next-token prediction meets full-sequence diffusion. *Proc. NeurIPS*, 2024a.
- 628
- 629 Ce Chen, Shaoli Huang, Xuelin Chen, Guangyi Chen, Xiaoguang Han, Kun Zhang, and Ming-
630 ming Gong. Ct4d: Consistent text-to-4d generation with animatable meshes. *arXiv preprint*
631 *arXiv:2408.08342*, 2024b.
- 632
- 633 Haoxin Chen, Menghan Xia, Yingqing He, Yong Zhang, Xiaodong Cun, Shaoshu Yang, Jinbo Xing,
634 Yaofang Liu, Qifeng Chen, Xintao Wang, Chao Weng, and Ying Shan. Videocrafter1: Open
635 diffusion models for high-quality video generation. *arXiv preprint arXiv:2310.19512*, 2023a.
- 636
- 637 Kevin Chen, Christopher B Choy, Manolis Savva, Angel X Chang, Thomas Funkhouser, and Silvio
638 Savarese. Text2Shape: Generating shapes from natural language by learning joint embeddings. In
639 *Proc. ACCV*, 2018.
- 640
- 641 Rui Chen, Yongwei Chen, Ningxin Jiao, and Kui Jia. Fantasia3D: Disentangling geometry and
642 appearance for high-quality text-to-3D content creation. *arXiv preprint arXiv:2303.13873*, 2023b.
- 643
- 644 Soon Yau Cheong, Duygu Ceylan, Armin Mustafa, Andrew Gilbert, and Chun-Hao Paul Huang.
645 Boosting camera motion control for video diffusion transformers. *arXiv preprint arXiv:2410.10802*,
646 2024.
- 647
- 648 Wen-Hsuan Chu, Lei Ke, and Katerina Fragkiadaki. Dreamscene4d: Dynamic multi-object scene
649 generation from monocular videos. *arXiv preprint arXiv:2405.02280*, 2024.
- 650
- 651 Team Cosmos. Cosmos world foundation model platform for physical ai. *arXiv preprint*
652 *arXiv:2501.03575*, 2025.
- 653
- 654 Tri Dao and Albert Gu. Transformers are ssms: Generalized models and efficient algorithms through
655 structured state space duality. *arXiv preprint arXiv:2405.21060*, 2024.

-
- Terrance DeVries, Miguel Angel Bautista, Nitish Srivastava, Graham W Taylor, and Joshua M Susskind. Unconstrained scene generation with locally conditioned radiance fields. In *Proc. ICCV*, 2021.
- Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *Proc. ICLR*, 2021.
- Qijun Feng, Zhen Xing, Zuxuan Wu, and Yu-Gang Jiang. FDGaussian: Fast Gaussian splatting from single image via geometric-aware diffusion model. *arXiv preprint arXiv:2403.10242*, 2024a.
- Yutao Feng, Yintong Shang, Xiang Feng, Lei Lan, Shandian Zhe, Tianjia Shao, Hongzhi Wu, Kun Zhou, Hao Su, Chenfanfu Jiang, et al. Elastogen: 4d generative elastodynamics. *arXiv preprint arXiv:2405.15056*, 2024b.
- Quankai Gao, Qiangeng Xu, Zhe Cao, Ben Mildenhall, Wenchao Ma, Le Chen, Danhang Tang, and Ulrich Neumann. Gaussianflow: Splatting Gaussian dynamics for 4D content creation. *arXiv preprint arXiv:2403.12365*, 2024a.
- Ruiqi Gao, Aleksander Holynski, Philipp Henzler, Arthur Brussee, Ricardo Martin-Brualla, Pratul Srinivasan, Jonathan T Barron, and Ben Poole. Cat3d: Create anything in 3d with multi-view diffusion models. In *Proc. NeurIPS*, 2024b.
- William Gao, Noam Aigerman, Thibault Groueix, Vova Kim, and Rana Hanocka. TextDeformer: Geometry manipulation using text guidance. In *SIGGRAPH*, 2023.
- Jiatao Gu, Alex Trevithick, Kai-En Lin, Joshua M Susskind, Christian Theobalt, Lingjie Liu, and Ravi Ramamoorthi. NerfDiff: Single-image view synthesis with NeRF-guided distillation from 3D-aware diffusion. In *Proc. ICML*, 2023.
- Zekai Gu, Rui Yan, Jiahao Lu, Peng Li, Zhiyang Dou, Chenyang Si, Zhen Dong, Qifeng Liu, Cheng Lin, Ziwei Liu, et al. Diffusion as shader: 3d-aware video diffusion for versatile video generation control. In *SIGGRAPH*, 2025.
- Junlin Han, Filippos Kokkinos, and Philip Torr. VFusion3D: Learning scalable 3D generative models from video diffusion models. *arXiv preprint arXiv:2403.12034*, 2024.
- Hao He, Yinghao Xu, Yuwei Guo, Gordon Wetzstein, Bo Dai, Hongsheng Li, and Ceyuan Yang. Cameractrl: Enabling camera control for text-to-video generation. *arXiv preprint arXiv:2404.02101*, 2024a.
- Hao He, Ceyuan Yang, Shanchuan Lin, Yinghao Xu, Meng Wei, Liangke Gui, Qi Zhao, Gordon Wetzstein, Lu Jiang, and Hongsheng Li. Cameractrl ii: Dynamic scene exploration via camera-controlled video diffusion models. *arXiv preprint arXiv:2503.10592*, 2025.
- Xianglong He, Junyi Chen, Sida Peng, Di Huang, Yangguang Li, Xiaoshui Huang, Chun Yuan, Wanli Ouyang, and Tong He. GVGEn: Text-to-3D generation with volumetric representation. *arXiv preprint arXiv:2403.12957*, 2024b.
- Lukas Höllein, Ang Cao, Andrew Owens, Justin Johnson, and Matthias Nießner. Text2room: Extracting textured 3d meshes from 2d text-to-image models. In *Proc. ICCV*, 2023.
- Lukas Höllein, Aljaž Božič, Norman Müller, David Novotny, Hung-Yu Tseng, Christian Richardt, Michael Zollhöfer, and Matthias Nießner. ViewDiff: 3D-consistent image generation with text-to-image models. In *Proc. CVPR*, 2024.
- Yicong Hong, Kai Zhang, Jiuxiang Gu, Sai Bi, Yang Zhou, Difan Liu, Feng Liu, Kalyan Sunkavalli, Trung Bui, and Hao Tan. LRM: Large reconstruction model for single image to 3D. In *Proc. ICLR*, 2024.
- Chen Hou, Guoqiang Wei, Yan Zeng, and Zhibo Chen. Training-free camera control for video generation. *arXiv preprint arXiv:2406.10126*, 2024.

-
- Ronghang Hu, Nikhila Ravi, Alexander C Berg, and Deepak Pathak. Worldsheet: Wrapping the world in a 3d sheet for view synthesis from a single image. In *Proc. ICCV*, 2021.
- Tao Hu, Haoyang Peng, Xiao Liu, and Yuwen Ma. Ex-4d: Extreme viewpoint 4d video synthesis via depth watertight mesh. *arXiv preprint arXiv:2506.05554*, 2025.
- Teng Hu, Jiangning Zhang, Ran Yi, Yating Wang, Hongrui Huang, Jieyu Weng, Yabiao Wang, and Lizhuang Ma. Motionmaster: Training-free camera motion transfer for video generation. *arXiv preprint arXiv:2404.15789*, 2024.
- Chun-Hao Paul Huang, Jae Shin Yoon, Hyeonho Jeong, Niloy Mitra, and Duygu Ceylan. Camera pose estimation emerging in video diffusion transformer, 2024a. URL <https://openreview.net/forum?id=lgf2LW7fOJ>.
- Jiahui Huang, Qunjie Zhou, Hesam Rabeti, Aleksandr Korovko, Huan Ling, Xuanchi Ren, Tianchang Shen, Jun Gao, Dmitry Slepichev, Chen-Hsuan Lin, Jiawei Ren, Kevin Xie, Joydeep Biswas, Laura Leal-Taixe, and Sanja Fidler. Vipe: Video pose engine for 3d geometric perception. In *NVIDIA Research Whitepapers*, 2025.
- Tianyu Huang, Yihan Zeng, Hui Li, Wangmeng Zuo, and Rynson WH Lau. Dreamphysics: Learning physical properties of dynamic 3d gaussians with video diffusion priors. *arXiv preprint arXiv:2406.01476*, 2024b.
- Ajay Jain, Ben Mildenhall, Jonathan T Barron, Pieter Abbeel, and Ben Poole. Zero-shot text-guided object generation with dream fields. In *Proc. CVPR*, 2022.
- Nikolay Jetchev. ClipMatrix: Text-controlled creation of 3D textured meshes. *arXiv preprint arXiv:2109.12922*, 2021.
- Lihan Jiang, Yucheng Mao, Linning Xu, Tao Lu, Kerui Ren, Yichen Jin, Xudong Xu, Mulin Yu, Jiangmiao Pang, Feng Zhao, et al. Anysplat: Feed-forward 3d gaussian splatting from unconstrained views. *arXiv preprint arXiv:2505.23716*, 2025a.
- Lutao Jiang and Lin Wang. Brightdreamer: Generic 3D Gaussian generative framework for fast text-to-3D synthesis. *arXiv preprint arXiv:2403.11273*, 2024.
- Yanqin Jiang, Chaohui Yu, Chenjie Cao, Fan Wang, Weiming Hu, and Jin Gao. Animate3d: Animating any 3d model with multi-view video diffusion. *arXiv preprint arXiv:2407.11398*, 2024a.
- Yanqin Jiang, Li Zhang, Jin Gao, Weimin Hu, and Yao Yao. Consistent4d: Consistent 360 $\{\backslash\deg\}$ dynamic object generation from monocular video. In *Proc. ICLR*, 2024b.
- Zeren Jiang, Chuanxia Zheng, Iro Laina, Diane Larlus, and Andrea Vedaldi. Geo4d: Leveraging video generators for geometric 4d scene reconstruction. *arXiv preprint arXiv:2504.07961*, 2025b.
- Wonjoon Jin, Qi Dai, Chong Luo, Seung-Hwan Baek, and Sunghyun Cho. Flovd: Optical flow meets video diffusion model for enhanced camera-controlled video synthesis. In *Proc. CVPR*, 2025.
- Xuan Ju, Weicai Ye, Quande Liu, Qiulin Wang, Xintao Wang, Pengfei Wan, Di Zhang, Kun Gai, and Qiang Xu. Fulldit: Multi-task video generative foundation model with full attention. *arXiv preprint arXiv:2503.19907*, 2025.
- Yash Kant, Aliaksandr Siarohin, Ziyi Wu, Michael Vasilkovsky, Guocheng Qian, Jian Ren, Riza Alp Guler, Bernard Ghanem, Sergey Tulyakov, and Igor Gilitschenski. Spad: Spatially aware multi-view diffusers. In *Proc. CVPR*, 2024.
- Yash Kant, Ethan Weber, Jin Kyu Kim, Rawal Khirodkar, Su Zhaoen, Julieta Martinez, Igor Gilitschenski, Shunsuke Saito, and Timur Bagautdinov. Pippo: High-resolution multi-view humans from a single image. In *Proc. CVPR*, 2025.
- Oren Katzir, Or Patashnik, Daniel Cohen-Or, and Dani Lischinski. Noise-free score distillation. In *Proc. ICLR*, 2024.
- Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3d gaussian splatting for real-time radiance field rendering. In *ACM TOG*, 2023.

756 Seung Wook Kim, Bradley Brown, Kangxue Yin, Karsten Kreis, Katja Schwarz, Daiqing Li, Robin
757 Rombach, Antonio Torralba, and Sanja Fidler. NeuralField-LDM: Scene generation with hierarchi-
758 cal latent diffusion models. In *Proc. CVPR*, 2023.

759 Diederik P Kingma and Max Welling. Auto-encoding variational bayes. *arXiv preprint*
760 *arXiv:1312.6114*, 2013.

761
762 Mert Kiray, Paul Uhlenbruck, Nassir Navab, and Benjamin Busam. Promptvfx: Text-driven fields for
763 open-world 3d gaussian animation. *arXiv preprint arXiv:2506.01091*, 2025.

764 Tobias Kirschstein, Javier Romero, Artem Sevastopolsky, Matthias Nießner, and Shunsuke Saito.
765 Avat3r: Large animatable gaussian reconstruction model for high-fidelity 3d head avatars. *arXiv*
766 *preprint arXiv:2502.20220*, 2025.

767
768 Arno Knapitsch, Jaesik Park, Qian-Yi Zhou, and Vladlen Koltun. Tanks and temples: Benchmarking
769 large-scale scene reconstruction. *ACM TOG*, 2017.

770
771 Zhengfei Kuang, Shengqu Cai, Hao He, Yinghao Xu, Hongsheng Li, Leonidas Guibas, and Gordon
772 Wetzstein. Collaborative video diffusion: Consistent multi-video generation with camera control.
773 In *Proc. NeurIPS*, 2024.

774
775 Kyungmin Lee, Kihyuk Sohn, and Jinwoo Shin. DreamFlow: High-quality text-to-3D generation by
776 approximating probability flow. In *Proc. ICLR*, 2024a.

777
778 Yao-Chih Lee, Yi-Ting Chen, Andrew Wang, Ting-Hsuan Liao, Brandon Y Feng, and Jia-Bin Huang.
779 Vividdream: Generating 3d scene with ambient dynamics. *arXiv preprint arXiv:2405.20334*,
780 2024b.

781
782 Guojun Lei, Chi Wang, Hong Li, Rong Zhang, Yikai Wang, and Weiwei Xu. Animateanything:
783 Consistent and controllable animation for video generation. *arXiv preprint arXiv:2411.10836*,
784 2024.

785
786 Guojun Lei, Chi Wang, Rong Zhang, Yikai Wang, Hong Li, and Weiwei Xu. Animateanything:
787 Consistent and controllable animation for video generation. In *Proc. CVPR*, 2025.

788
789 Bing Li, Cheng Zheng, Wenxuan Zhu, Jinjie Mai, Biao Zhang, Peter Wonka, and Bernard Ghanem.
790 Vivid-zoo: Multi-view video generation with diffusion model. *arXiv preprint arXiv:2406.08659*,
791 2024a.

792
793 Jiahao Li, Hao Tan, Kai Zhang, Zexiang Xu, Fujun Luan, Yinghao Xu, Yicong Hong, Kalyan
794 Sunkavalli, Greg Shakhnarovich, and Sai Bi. Instant3D: Fast text-to-3D with sparse-view genera-
795 tion and large reconstruction model. In *Proc. ICLR*, 2024b.

796
797 Jinwei Li, Huan-ang Gao, Wenyi Li, Haohan Chi, Chenyu Liu, Chenxi Du, Yiqian Liu, Mingju Gao,
798 Guiyu Zhang, Zongzheng Zhang, et al. Fb-4d: Spatial-temporal coherent dynamic 3d content
799 generation with feature banks. *arXiv preprint arXiv:2503.20784*, 2025a.

800
801 Longfei Li, Zhiwen Fan, Wenyan Cong, Xinhang Liu, Yuyang Yin, Matt Foutter, Panwang Pan,
802 Chenyu You, Yue Wang, Zhangyang Wang, et al. Martian world models: Controllable video
803 synthesis with physically accurate 3d reconstructions. *arXiv preprint arXiv:2507.07978*, 2025b.

804
805 Renjie Li, Panwang Pan, Bangbang Yang, Dejia Xu, Shijie Zhou, Xuanyang Zhang, Zeming Li,
806 Achuta Kadambi, Zhangyang Wang, and Zhiwen Fan. 4k4dgen: Panoramic 4d generation at 4k
807 resolution. *arXiv preprint arXiv:2406.13527*, 2024c.

808
809 Teng Li, Guangcong Zheng, Rui Jiang, Shuigen Zhan, Tao Wu, Yehao Lu, Yining Lin, Chuanyun
Deng, Yapan Xiong, Min Chen, et al. Realcam-i2v: Real-world image-to-video generation with
interactive complex camera control. *arXiv preprint arXiv:2502.10059*, 2025c.

Tianye Li, Mira Slavcheva, Michael Zollhoefer, Simon Green, Christoph Lassner, Changil Kim,
Tanner Schmidt, Steven Lovegrove, Michael Goesele, Richard Newcombe, et al. Neural 3d video
synthesis from multi-view video. In *Proc. CVPR*, 2022.

-
- Zhiqi Li, Yiming Chen, and Peidong Liu. Dreammesh4d: Video-to-4d generation with sparse-controlled gaussian-mesh hybrid representation. *arXiv preprint arXiv:2410.06756*, 2024d.
- Zhiqi Li, Yiming Chen, Lingzhe Zhao, and Peidong Liu. Controllable text-to-3D generation via surface-aligned Gaussian splatting. *arXiv preprint arXiv:2403.09981*, 2024e.
- Hanwen Liang, Yuyang Yin, Dejia Xu, Hanxue Liang, Zhangyang Wang, Konstantinos N Plataniotis, Yao Zhao, and Yunchao Wei. Diffusion4d: Fast spatial-temporal consistent 4d generation via video diffusion models. *arXiv preprint arXiv:2405.16645*, 2024.
- Hanwen Liang, Junli Cao, Vidit Goel, Guocheng Qian, Sergei Korolev, Demetri Terzopoulos, Konstantinos N Plataniotis, Sergey Tulyakov, and Jian Ren. Wonderland: Navigating 3d scenes from a single image. *Proc. CVPR*, 2025a.
- Hanxue Liang, Jiawei Ren, Ashkan Mirzaei, Antonio Torralba, Ziwei Liu, Igor Gilitschenski, Sanja Fidler, Cengiz Oztireli, Huan Ling, Zan Gojcic, and Jiahui Huang. Feed-forward bullet-time reconstruction of dynamic scenes from monocular videos. *Proc. NeurIPS*, 2025b.
- Yixun Liang, Xin Yang, Jiantao Lin, Haodong Li, Xiaogang Xu, and Yingcong Chen. Lucid-dreamer: Towards high-fidelity text-to-3D generation via interval score matching. *arXiv preprint arXiv:2311.11284*, 2023.
- Chen-Hsuan Lin, Jun Gao, Luming Tang, Towaki Takikawa, Xiaohui Zeng, Xun Huang, Karsten Kreis, Sanja Fidler, Ming-Yu Liu, and Tsung-Yi Lin. Magic3D: High-resolution text-to-3D content creation. In *Proc. CVPR*, 2023a.
- Chenguo Lin, Yuchen Lin, Panwang Pan, Yifan Yu, Honglei Yan, Katerina Fragkiadaki, and Yadong Mu. Movies: Motion-aware 4d dynamic view synthesis in one second. *arXiv preprint arXiv:2507.10065*, 2025.
- Jiajing Lin, Zhenzhong Wang, Yongjie Hou, Yuzhou Tang, and Min Jiang. Phy124: Fast physics-driven 4d content generation from a single image. *arXiv preprint arXiv:2409.07179*, 2024.
- Yukang Lin, Haonan Han, Chaoqun Gong, Zunnan Xu, Yachao Zhang, and Xiu Li. Consistent123: One image to highly consistent 3D asset using case-aware diffusion priors. In *arXiv preprint arXiv:2309.17261*, 2023b.
- Huan Ling, Seung Wook Kim, Antonio Torralba, Sanja Fidler, and Karsten Kreis. Align your gaussians: Text-to-4d with dynamic 3d gaussians and composed diffusion models. In *Proc. CVPR*, 2024a.
- Lu Ling, Yichen Sheng, Zhi Tu, Wentian Zhao, Cheng Xin, Kun Wan, Lantao Yu, Qianyu Guo, Zixun Yu, Yawen Lu, et al. D13dv-10k: A large-scale scene dataset for deep learning-based 3d vision. In *Proc. CVPR*, 2024b.
- Pengyang Ling, Jiazi Bu, Pan Zhang, Xiaoyi Dong, Yuhang Zang, Tong Wu, Huaian Chen, Jiaqi Wang, and Yi Jin. Motionclone: Training-free motion cloning for controllable video generation. *arXiv preprint arXiv:2406.05338*, 2024c.
- Lijuan Liu, Wenfa Li, Dongbo Zhang, Shuo Wang, and Shaohui Jiao. Idcnet: Guided video diffusion for metric-consistent rgbd scene generation with precise camera control. *arXiv preprint arXiv:2508.04147*, 2025.
- Pengkun Liu, Yikai Wang, Fuchun Sun, Jiafang Li, Hang Xiao, Hongxiang Xue, and Xinzhou Wang. Isotropic3D: Image-to-3D generation based on a single clip embedding. *arXiv preprint arXiv:2403.10395*, 2024a.
- Ruoshi Liu, Rundi Wu, Basile Van Hoorick, Pavel Tokmakov, Sergey Zakharov, and Carl Vondrick. Zero-1-to-3: Zero-shot one image to 3D object. In *Proc. ICCV*, 2023.
- Xian Liu, Xiaohang Zhan, Jiayang Tang, Ying Shan, Gang Zeng, Dahua Lin, Xihui Liu, and Ziwei Liu. HumanGaussian: Text-driven 3D human generation with Gaussian splatting. In *Proc. CVPR*, 2024b.

-
- Yuan Liu, Cheng Lin, Zijiao Zeng, Xiaoxiao Long, Lingjie Liu, Taku Komura, and Wenping Wang. SyncDreamer: Generating multiview-consistent images from a single-view image. In *Proc. ICLR*, 2024c.
- Xiaoxiao Long, Yuan-Chen Guo, Cheng Lin, Yuan Liu, Zhiyang Dou, Lingjie Liu, Yuexin Ma, Song-Hai Zhang, Marc Habermann, Christian Theobalt, et al. Wonder3D: Single image to 3D using cross-domain diffusion. In *Proc. CVPR*, 2024.
- Yifan Lu, Xuanchi Ren, Jiawei Yang, Tianchang Shen, Zhangjie Wu, Jun Gao, Yue Wang, Siheng Chen, Mike Chen, Sanja Fidler, et al. Infinicube: Unbounded and controllable dynamic 3d driving scene generation with world-guided video models. *arXiv preprint arXiv:2412.03934*, 2024.
- Yue Ma, Kunyu Feng, Xinhua Zhang, Hongyu Liu, David Junhao Zhang, Jinbo Xing, Yinhan Zhang, Ayden Yang, Zeyu Wang, and Qifeng Chen. Follow-your-creation: Empowering 4d creation through video inpainting. *arXiv preprint arXiv:2506.04590*, 2025.
- Jinjie Mai, Wenxuan Zhu, Haozhe Liu, Bing Li, Cheng Zheng, Jürgen Schmidhuber, and Bernard Ghanem. Can video diffusion model reconstruct 4d geometry? *arXiv preprint arXiv:2503.21082*, 2025.
- Qiaowei Miao, Yawei Luo, and Yi Yang. Pla4d: Pixel-level alignments for text-to-4d gaussian splatting. *arXiv preprint arXiv:2405.19957*, 2024.
- Ben Mildenhall, Pratul P Srinivasan, Matthew Tancik, Jonathan T Barron, Ravi Ramamoorthi, and Ren Ng. Nerf: Representing scenes as neural radiance fields for view synthesis. In *Proc. ECCV*, 2020.
- Nicolas Moenne-Loccoz, Ashkan Mirzaei, Or Perel, Riccardo de Lutio, Janick Martinez Esturo, Gavriel State, Sanja Fidler, Nicholas Sharp, and Zan Gojcic. 3d gaussian ray tracing: Fast tracing of particle scenes. *ACM Transactions on Graphics and SIGGRAPH Asia*, 2024.
- OpenAI. Chatgpt (gpt-5), 2025. Large language model. Available at <https://chat.openai.com/>.
- Roy Or-El, Xuan Luo, Mengyi Shan, Eli Shechtman, Jeong Joon Park, and Ira Kemelmacher-Shlizerman. StyleSDF: High-resolution 3D-consistent image and geometry generation. In *Proc. CVPR*, 2022.
- Zijie Pan, Zeyu Yang, Xiatian Zhu, and Li Zhang. Fast dynamic 3d object generation from a single-view video. *arXiv preprint arXiv:2401.08742*, 2024.
- Jangho Park, Taesung Kwon, and Jong Chul Ye. Zero4d: Training-free 4d video generation from single video using off-the-shelf video diffusion. *arXiv preprint arXiv:2503.22622*, 2025.
- Ben Poole, Ajay Jain, Jonathan T. Barron, and Ben Mildenhall. DreamFusion: Text-to-3D using 2D diffusion. In *Proc. ICLR*, 2023.
- Guocheng Qian, Junli Cao, Aliaksandr Siarohin, Yash Kant, Chaoyang Wang, Michael Vasilkovsky, Hsin-Ying Lee, Yuwei Fang, Ivan Skorokhodov, Peiye Zhuang, et al. Atom: Amortized text-to-mesh using 2d diffusion. *arXiv preprint arXiv:2402.00867*, 2024a.
- Guocheng Qian, Jinjie Mai, Abdullah Hamdi, Jian Ren, Aliaksandr Siarohin, Bing Li, Hsin-Ying Lee, Ivan Skorokhodov, Peter Wonka, Sergey Tulyakov, et al. Magic123: One image to high-quality 3D object generation using both 2D and 3D diffusion priors. In *Proc. ICLR*, 2024b.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *Proc. ICML*, 2021.
- Jiawei Ren, Liang Pan, Jiaxiang Tang, Chi Zhang, Ang Cao, Gang Zeng, and Ziwei Liu. DreamGaussian4D: Generative 4D Gaussian splatting. *arXiv preprint arXiv:2312.17142*, 2023.
- Jiawei Ren, Kevin Xie, Ashkan Mirzaei, Hanxue Liang, Xiaohui Zeng, Karsten Kreis, Ziwei Liu, Antonio Torralba, Sanja Fidler, Seung Wook Kim, et al. L4gm: Large 4d gaussian reconstruction model. In *Proc. NeurIPS*, 2024a.

918 Xuanchi Ren, Jiahui Huang, Xiaohui Zeng, Ken Museth, Sanja Fidler, and Francis Williams. Xcube:
919 Large-scale 3d generative modeling using sparse voxel hierarchies. In *Proc. CVPR*, 2024b.

920

921 Xuanchi Ren, Yifan Lu, Hanxue Liang, Zhangjie Wu, Huan Ling, Mike Chen, Sanja Fidler, Francis
922 Williams, and Jiahui Huang. Scube: Instant large-scale scene reconstruction using voxplats. *Proc.*
923 *NeurIPS*, 2024c.

924 Xuanchi Ren, Tianchang Shen, Jiahui Huang, Huan Ling, Yifan Lu, Merlin Nimier-David, Thomas
925 Müller, Alexander Keller, Sanja Fidler, and Jun Gao. Gen3c: 3d-informed world-consistent video
926 generation with precise camera control. *Proc. CVPR*, 2025.

927

928 Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-
929 resolution image synthesis with latent diffusion models. In *Proc. CVPR*, 2022.

930 Aditya Sanghi, Hang Chu, Joseph G Lambourne, Ye Wang, Chin-Yi Cheng, Marco Fumero, and
931 Kamal Rahimi Malekshan. CLIP-Forge: Towards zero-shot text-to-shape generation. In *Proc.*
932 *CVPR*, 2022.

933

934 Kyle Sargent, Zizhang Li, Tanmay Shah, Charles Herrmann, Hong-Xing Yu, Yunzhi Zhang, Eric Ryan
935 Chan, Dmitry Lagun, Li Fei-Fei, Deqing Sun, et al. Zeronvs: Zero-shot 360-degree view synthesis
936 from a single image. In *Proc. CVPR*, 2024.

937 Manuel-Andreas Schneider, Lukas Höllein, and Matthias Nießner. Worldexplorer: Towards generating
938 fully navigable 3d scenes. *arXiv preprint arXiv:2506.01799*, 2025.

939

940 Katja Schwarz, Axel Sauer, Michael Niemeyer, Yiyi Liao, and Andreas Geiger. VoxGRAF: Fast
941 3D-aware image synthesis with sparse voxel grids. In *Proc. NeurIPS*, 2022.

942 Katja Schwarz, Norman Mueller, and Peter Kotschieder. Generative gaussian splatting: Generating
943 3d scenes with video diffusion priors. *arXiv preprint arXiv:2503.13272*, 2025.

944

945 Junyoung Seo, Jisang Han, Jaewoo Jung, Siyoon Jin, Jounghbin Lee, Takuya Narihira, Kazumi Fukuda,
946 Takashi Shibuya, Donghoon Ahn, Shoukang Hu, et al. Vid-camedit: Video camera trajectory
947 editing with generative rendering from estimated geometry. *arXiv preprint arXiv:2506.13697*,
948 2025.

949 Yichun Shi, Peng Wang, Jianglong Ye, Long Mai, Kejie Li, and Xiao Yang. MVDream: Multi-view
950 diffusion for 3D generation. In *Proc. ICLR*, 2024.

951

952 Jaidev Shriram, Alex Trevithick, Lingjie Liu, and Ravi Ramamoorthi. Realmdreamer: Text-driven 3d
953 scene generation with inpainting and depth diffusion. *arXiv preprint arXiv:2404.07199*, 2024.

954 Karen Simonyan and Andrew Zisserman. Very deep convolutional networks for large-scale image
955 recognition. *arXiv preprint arXiv:1409.1556*, 2014.

956

957 Uriel Singer, Shelly Sheynin, Adam Polyak, Oron Ashual, Iurii Makarov, Filippos Kokkinos, Naman
958 Goyal, Andrea Vedaldi, Devi Parikh, Justin Johnson, et al. Text-to-4d dynamic scene generation.
959 In *Proc. ICML*, 2023.

960 Jingxiang Sun, Bo Zhang, Ruizhi Shao, Lizhen Wang, Wen Liu, Zhenda Xie, and Yebin Liu.
961 DreamCraft3D: Hierarchical 3D generation with bootstrapped diffusion prior. In *Proc. ICLR*,
962 2024a.

963

964 Keqiang Sun, Dor Litvak, Yunzhi Zhang, Hongsheng Li, Jiajun Wu, and Shangzhe Wu. Ponymation:
965 Learning articulated 3d animal motions from unlabeled online videos. In *Proc. ECCV*, 2024b.

966

967 Qi Sun, Zhiyang Guo, Ziyu Wan, Jing Nathan Yan, Shengming Yin, Wengang Zhou, Jing Liao, and
968 Houqiang Li. Eg4d: Explicit generation of 4d object without score distillation. *arXiv preprint*
969 *arXiv:2405.18132*, 2024c.

970 Wenqiang Sun, Shuo Chen, Fangfu Liu, Zilong Chen, Yueqi Duan, Jun Zhang, and Yikai Wang.
971 Dimensionx: Create any 3d and 4d scenes from a single image with controllable video diffusion.
arXiv preprint arXiv:2411.04928, 2024d.

972 Stanislaw Szymanowicz, Christian Rupprecht, and Andrea Vedaldi. Viewset diffusion:(0-) image-
973 conditioned 3d generative models from 2d data. In *Proc. ICCV*, 2023.

974

975 Stanislaw Szymanowicz, Christian Rupprecht, and Andrea Vedaldi. Splatter image: Ultra-fast
976 single-view 3D reconstruction. In *Proc. CVPR*, 2024.

977

978 Stanislaw Szymanowicz, Eldar Insafutdinov, Chuanxia Zheng, Dylan Campbell, João F Henriques,
979 Christian Rupprecht, and Andrea Vedaldi. Flash3d: Feed-forward generalisable 3d scene recon-
980 struction from a single image. *Proc. 3DV*, 2025a.

981

982 Stanislaw Szymanowicz, Jason Y Zhang, Pratul Srinivasan, Ruiqi Gao, Arthur Brussee, Aleksander
983 Holynski, Ricardo Martin-Brualla, Jonathan T Barron, and Philipp Henzler. Bolt3d: Generating 3d
984 scenes in seconds. *arXiv preprint arXiv:2503.14445*, 2025b.

985

986 Jiaxiang Tang, Zhaoxi Chen, Xiaokang Chen, Tengfei Wang, Gang Zeng, and Ziwei Liu. LGM:
987 Large multi-view gaussian model for high-resolution 3d content creation. In *Proc. ECCV*, 2024a.

988

989 Jiaxiang Tang, Zhaoxi Chen, Xiaokang Chen, Tengfei Wang, Gang Zeng, and Ziwei Liu. LGM:
990 Large multi-view gaussian model for high-resolution 3d content creation. *Proc. ECCV*, 2024b.

991

992 Junshu Tang, Tengfei Wang, Bo Zhang, Ting Zhang, Ran Yi, Lizhuang Ma, and Dong Chen.
993 Make-it-3D: High-fidelity 3D creation from a single image with diffusion prior. *arXiv preprint*
994 *arXiv:2303.14184*, 2023a.

995

996 Shitao Tang, Fuyang Zhang, Jiacheng Chen, Peng Wang, and Yasutaka Furukawa. Mvdifusion: En-
997 abling holistic multi-view image generation with correspondence-aware diffusion. *Proc. NeurIPS*,
998 2023b.

999

1000 Zhenggang Tang, Peiye Zhuang, Chaoyang Wang, Aliaksandr Siarohin, Yash Kant, Alexander
1001 Schwing, Sergey Tulyakov, and Hsin-Ying Lee. Pixel-aligned multi-view generation with depth
1002 guided decoder. *arXiv preprint arXiv:2408.14016*, 2024c.

1003

1004 Ayush Tewari, Tianwei Yin, George Cazenavette, Semon Rezhikov, Josh Tenenbaum, Frédo Durand,
1005 Bill Freeman, and Vincent Sitzmann. Diffusion with forward models: Solving stochastic inverse
1006 problems without direct supervision. In *Proc. NeurIPS*, 2023.

1007

1008 Dmitry Tochilkin, David Pankratz, Zexiang Liu, Zixuan Huang, Adam Letts, Yangguang Li, Ding
1009 Liang, Christian Laforte, Varun Jampani, and Yan-Pei Cao. Triposr: Fast 3D object reconstruction
1010 from a single image. *arXiv preprint arXiv:2403.02151*, 2024.

1011

1012 Lukas Uzolas, Elmar Eisemann, and Petr Kellnhofer. Motiondreamer: Zero-shot 3d mesh animation
1013 from video diffusion models. *arXiv preprint arXiv:2405.20155*, 2024.

1014

1015 Basile Van Hoorick, Rundi Wu, Ege Ozguroglu, Kyle Sargent, Ruoshi Liu, Pavel Tokmakov, Achal
1016 Dave, Changxi Zheng, and Carl Vondrick. Generative camera dolly: Extreme monocular dynamic
1017 novel view synthesis. In *Proc. ECCV*, 2024.

1018

1019 Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz
1020 Kaiser, and Illia Polosukhin. Attention is all you need. In *Proc. NeurIPS*, 2017.

1021

1022 Vikram Voleti, Chun-Han Yao, Mark Boss, Adam Letts, David Pankratz, Dmitry Tochilkin, Chris-
1023 tian Laforte, Robin Rombach, and Varun Jampani. SV3D: Novel multi-view synthesis and 3D
1024 generation from a single image using latent video diffusion. *arXiv preprint arXiv:2403.12008*,
1025 2024.

Team Wan. Wan: Open and advanced large-scale video generative models. *arXiv preprint*
arXiv:2503.20314, 2025.

Ziyu Wan, Despoina Paschalidou, Ian Huang, Hongyu Liu, Bokui Shen, Xiaoyu Xiang, Jing Liao,
and Leonidas Guibas. CAD: Photorealistic 3D generation via adversarial distillation. In *Proc.*
CVPR, 2024.

Can Wang, Menglei Chai, Mingming He, Dongdong Chen, and Jing Liao. Clip-NeRF: Text-and-image
driven manipulation of neural radiance fields. In *Proc. CVPR*, 2022.

1026 Chaoyang Wang, Peiye Zhuang, Tuan Duc Ngo, Willi Menapace, Aliaksandr Siarohin, Michael
1027 Vasilkovsky, Ivan Skorokhodov, Sergey Tulyakov, Peter Wonka, and Hsin-Ying Lee. 4real-video:
1028 Learning generalizable photo-realistic 4d video diffusion. *Proc. CVPR*, 2025a.

1029 Haochen Wang, Xiaodan Du, Jiahao Li, Raymond A Yeh, and Greg Shakhnarovich. Score Jacobian
1030 chaining: Lifting pretrained 2d diffusion models for 3D generation. In *Proc. CVPR*, 2023a.

1032 Peng Wang and Yichun Shi. Imagedream: Image-prompt multi-view diffusion for 3d generation.
1033 *arXiv preprint arXiv:2312.02201*, 2023.

1034 Qinghe Wang, Yawen Luo, Xiaoyu Shi, Xu Jia, Huchuan Lu, Tianfan Xue, Xintao Wang, Pengfei
1035 Wan, Di Zhang, and Kun Gai. Cinemaster: A 3d-aware and controllable framework for cinematic
1036 text-to-video generation. In *SIGGRAPH*, 2025b.

1038 Ruicheng Wang, Sicheng Xu, Cassie Dai, Jianfeng Xiang, Yu Deng, Xin Tong, and Jiaolong Yang.
1039 Moge: Unlocking accurate monocular geometry estimation for open-domain images with optimal
1040 training supervision, 2024a. URL <https://arxiv.org/abs/2410.19115>.

1042 Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jerome Revaud. Dust3r:
1043 Geometric 3d vision made easy. In *Proc. CVPR*, 2024b.

1044 Xi Wang, Robin Courant, Marc Christie, and Vicky Kalogeiton. Akira: Augmentation kit on rays for
1045 optical video generation. In *Proc. CVPR*, 2025c.

1047 Xiaodong Wang, Zhirong Wu, and Peixi Peng. Longdwm: Cross-granularity distillation for building
1048 a long-term driving world model. *arXiv preprint arXiv:2506.01546*, 2025d.

1049 Yikai Wang, Xinzhou Wang, Zilong Chen, Zhengyi Wang, Fuchun Sun, and Jun Zhu. Vidu4d: Single
1050 generated video to high-fidelity 4d reconstruction with dynamic gaussian surfels. *arXiv preprint*
1051 *arXiv:2405.16822*, 2024c.

1053 Yikai Wang, Guangce Liu, Xinzhou Wang, Zilong Chen, Jiafang Li, Xin Liang, Fuchun Sun, and
1054 Jun Zhu. Video4dgen: Enhancing video and 4d generation through mutual optimization. *arXiv*
1055 *preprint arXiv:2504.04153*, 2025e.

1056 Yizhi Wang, Mingrui Zhao, Ali Mahdavi-Amiri, and Hao Zhang. Act-r: Adaptive camera trajectories
1057 for single view 3d reconstruction. *arXiv preprint arXiv:2505.08239*, 2025f.

1059 Yuelei Wang, Jian Zhang, Pengtao Jiang, Hao Zhang, Jinwei Chen, and Bo Li. Cpa: Camera-
1060 pose-awareness diffusion transformer for video generation. *arXiv preprint arXiv:2412.01429*,
1061 2024d.

1062 Zhengyi Wang, Cheng Lu, Yikai Wang, Fan Bao, Chongxuan Li, Hang Su, and Jun Zhu. Prolific-
1063 Dreamer: High-fidelity and diverse text-to-3D generation with variational score distillation. In
1064 *Proc. NeurIPS*, 2023b.

1066 Zhouxia Wang, Ziyang Yuan, Xintao Wang, Tianshui Chen, Menghan Xia, Ping Luo, and Yin Shan.
1067 Motionctrl: A unified and flexible motion controller for video generation. In *SIGGRAPH*, 2024e.

1068 Daniel Watson, William Chan, Ricardo Martin-Brualla, Jonathan Ho, Andrea Tagliasacchi, and
1069 Mohammad Norouzi. Novel view synthesis with diffusion models. *Proc. ICLR*, 2023.

1070 Daniel Watson, Saurabh Saxena, Lala Li, Andrea Tagliasacchi, and David J Fleet. Controlling space
1071 and time with diffusion models. *arXiv preprint arXiv:2407.07860*, 2024.

1073 Qi Wu, Janick Martinez Esturo, Ashkan Mirzaei, Nicolas Moenne-Loccoz, and Zan Gojcic. 3dgut:
1074 Enabling distorted cameras and secondary rays in gaussian splatting. In *Proc. CVPR*, 2025a.

1075 Rundui Wu, Ruiqi Gao, Ben Poole, Alex Trevithick, Changxi Zheng, Jonathan T Barron, and Alek-
1076 sander Holynski. Cat4d: Create anything in 4d with multi-view video diffusion models. *Proc.*
1077 *CVPR*, 2025b.

1079 Tong Wu, Shuai Yang, Ryan Po, Yinghao Xu, Ziwei Liu, Dahua Lin, and Gordon Wetzstein. Video
world models with long-term spatial memory. *arXiv preprint arXiv:2506.05284*, 2025c.

1080 Zijie Wu, Chaohui Yu, Yanqin Jiang, Chenjie Cao, Fan Wang, and Xiang Bai. Sc4d: Sparse-controlled
1081 video-to-4d generation and motion transfer. *arXiv preprint arXiv:2404.03736*, 2024.

1082

1083 Zeqi Xiao, Yifan Zhou, Shuai Yang, and Xingang Pan. Video diffusion models are training-free
1084 motion interpreter and controller. *arXiv preprint arXiv:2405.14864*, 2024.

1085

1086 Kevin Xie, Jonathan Lorraine, Tianshi Cao, Jun Gao, James Lucas, Antonio Torralba, Sanja Fidler,
1087 and Xiaohui Zeng. LATTE3D: Large-scale amortized text-to-enhanced3D synthesis. In *Proc.
1088 ECCV*, 2024a.

1089

1090 Yiming Xie, Chun-Han Yao, Vikram Voleti, Huaizu Jiang, and Varun Jampani. Sv4d: Dynamic 3d
1091 content generation with multi-frame and multi-view consistency. *arXiv preprint arXiv:2407.17470*,
1092 2024b.

1093

1094 Jinbo Xing, Long Mai, Cusuh Ham, Jiahui Huang, Aniruddha Mahapatra, Chi-Wing Fu, Tien-Tsin
1095 Wong, and Feng Liu. Motioncanvas: Cinematic shot design with controllable image-to-video
1096 generation. In *SIGGRAPH*, 2025.

1097

1098 Dejia Xu, Yifan Jiang, Chen Huang, Liangchen Song, Thorsten Gernoth, Liangliang Cao, Zhangyang
1099 Wang, and Hao Tang. Cavia: Camera-controllable multi-view video diffusion with view-integrated
1100 attention. *arXiv preprint arXiv:2410.10774*, 2024a.

1101

1102 Dejia Xu, Hanwen Liang, Neel P Bhatt, Hezhen Hu, Hanxue Liang, Konstantinos N Plataniotis,
1103 and Zhangyang Wang. Comp4d: Llm-guided compositional 4d scene generation. *arXiv preprint
1104 arXiv:2403.16993*, 2024b.

1105

1106 Dejia Xu, Weili Nie, Chao Liu, Sifei Liu, Jan Kautz, Zhangyang Wang, and Arash Vahdat. Camco:
1107 Camera-controllable 3d-consistent image-to-video generation. *arXiv preprint arXiv:2406.02509*,
1108 2024c.

1109

1110 Yinghao Xu, Zifan Shi, Wang Yifan, Hansheng Chen, Ceyuan Yang, Sida Peng, Yujun Shen, and
1111 Gordon Wetzstein. GRM: Large Gaussian reconstruction model for efficient 3D reconstruction
1112 and generation. In *Proc. ECCV*, 2024d.

1113

1114 Yinghao Xu, Hao Tan, Fujun Luan, Sai Bi, Peng Wang, Jiahao Li, Zifan Shi, Kalyan Sunkavalli,
1115 Gordon Wetzstein, Zexiang Xu, et al. DMV3D: Denoising multi-view diffusion using 3D large
1116 reconstruction model. In *Proc. ICLR*, 2024e.

1117

1118 Zhen Xu, Zhengqin Li, Zhao Dong, Xiaowei Zhou, Richard Newcombe, and Zhaoyang Lv.
1119 4dgt: Learning a 4d gaussian transformer using real-world monocular videos. *arXiv preprint
1120 arXiv:2506.08015*, 2025.

1121

1122 Zhongcong Xu, Jianfeng Zhang, Jun Hao Liew, Wenqing Zhang, Song Bai, Jiashi Feng, and
1123 Mike Zheng Shou. Pv3d: A 3d generative model for portrait video generation. In *Proc. ICLR*,
1124 2023.

1125

1126 Yunzhi Yan, Zhen Xu, Haotong Lin, Haian Jin, Haoyu Guo, Yida Wang, Kun Zhan, Xianpeng Lang,
1127 Hujun Bao, Xiaowei Zhou, et al. Streetcrafter: Street view synthesis with controllable video
1128 diffusion models. In *Proc. CVPR*, 2025.

1129

1130 Qitong Yang, Mingtao Feng, Zijie Wu, Shijie Sun, Weisheng Dong, Yaonan Wang, and Ajmal Mian.
1131 Beyond skeletons: Integrative latent mapping for coherent 4d sequence generation. *arXiv preprint
1132 arXiv:2403.13238*, 2024a.

1133

1134 Zeyu Yang, Zijie Pan, Chun Gu, and Li Zhang. Diffusion²: Dynamic 3d content generation via score
1135 composition of orthogonal diffusion models. *arXiv preprint 2404.02148*, 2024b.

1136

1137 Zhongqi Yang, Wenhang Ge, Yuqi Li, Jiaqi Chen, Haoyuan Li, Mengyin An, Fei Kang, Hua Xue,
1138 Baixin Xu, Yuyang Yin, et al. Matrix-3d: Omnidirectional explorable 3d world generation. *arXiv
1139 preprint arXiv:2508.08086*, 2025.

1140

1141 Junliang Ye, Fangfu Liu, Qixiu Li, Zhengyi Wang, Yikai Wang, Xinzhou Wang, Yueqi Duan,
1142 and Jun Zhu. DreamReward: Text-to-3D generation with human preference. *arXiv preprint
1143 arXiv:2403.14613*, 2024.

1134 Vickie Ye, Ruilong Li, Justin Kerr, Matias Turkulainen, Brent Yi, Zhuoyang Pan, Otto Seiskari,
1135 Jianbo Ye, Jeffrey Hu, Matthew Tancik, and Angjoo Kanazawa. gsplat: An open-source library for
1136 gaussian splatting. *JMLR*, 2025.

1137 Yuyang Yin, Dejia Xu, Zhangyang Wang, Yao Zhao, and Yunchao Wei. 4dgen: Grounded 4d content
1138 generation with spatial-temporal consistency. *arXiv preprint arXiv:2312.17225*, 2023.

1140 Paul Yoo, Jiaxian Guo, Yutaka Matsuo, and Shixiang Shane Gu. DreamSparse: Escaping from Plato’s
1141 cave with 2D diffusion model given sparse views. In *arXiv preprint arXiv:2306.03414*, 2023.

1142 Heng Yu, Chaoyang Wang, Peiye Zhuang, Willi Menapace, Aliaksandr Siarohin, Junli Cao, Laszlo A
1143 Jeni, Sergey Tulyakov, and Hsin-Ying Lee. 4real: Towards photorealistic 4d scene generation via
1144 video diffusion models. In *Proc. NeurIPS*, 2024a.

1146 Mark YU, Wenbo Hu, Jinbo Xing, and Ying Shan. Trajectorycrafter: Redirecting camera trajectory
1147 for monocular videos via diffusion models. *Proc. ICCV*, 2025.

1149 Wangbo Yu, Jinbo Xing, Li Yuan, Wenbo Hu, Xiaoyu Li, Zhipeng Huang, Xiangjun Gao, Tien-
1150 Tsin Wong, Ying Shan, and Yonghong Tian. Viewcrafter: Taming video diffusion models for
1151 high-fidelity novel view synthesis. *arXiv preprint arXiv:2409.02048*, 2024b.

1152 Xin Yu, Yuan-Chen Guo, Yangguang Li, Ding Liang, Song-Hai Zhang, and Xiaojuan Qi. Text-to-3D
1153 with classifier score distillation. *arXiv preprint arXiv:2310.19415*, 2023.

1155 Yu Yuan, Xijun Wang, Yichen Sheng, Prateek Chennuri, Xingguang Zhang, and Stanley Chan.
1156 Generative photography: Scene-consistent camera control for realistic text-to-image synthesis. In
1157 *Proc. CVPR*, 2025.

1158 Yu-Jie Yuan, Leif Kobbelt, Jiwen Liu, Yuan Zhang, Pengfei Wan, Yu-Kun Lai, and Lin Gao. 4dynamic:
1159 Text-to-4d generation with hybrid priors. *arXiv preprint arXiv:2407.12684*, 2024.

1161 Bohan Zeng, Ling Yang, Siyu Li, Jiaming Liu, Zixiang Zhang, Juanxi Tian, Kaixin Zhu, Yongzhen
1162 Guo, Fu-Yun Wang, Minkai Xu, et al. Trans4d: Realistic geometry-aware transition for composi-
1163 tional text-to-4d synthesis. *arXiv preprint arXiv:2410.07155*, 2024a.

1164 Yifei Zeng, Yanqin Jiang, Siyu Zhu, Yuanxun Lu, Youtian Lin, Hao Zhu, Weiming Hu, Xun
1165 Cao, and Yao Yao. Stag4d: Spatial-temporal anchored generative 4d gaussians. *arXiv preprint*
1166 *arXiv:2403.14939*, 2024b.

1168 Bowen Zhang, Tianyu Yang, Yu Li, Lei Zhang, and Xi Zhao. Compress3D: a compressed latent
1169 space for 3D generation from a single image. *arXiv preprint arXiv:2403.13524*, 2024a.

1171 David Junhao Zhang, Roni Paiss, Shiran Zada, Nikhil Karnad, David E Jacobs, Yael Pritch, Inbar
1172 Mosseri, Mike Zheng Shou, Neal Wadhwa, and Nataniel Ruiz. Recapture: Generative video camera
1173 controls for user-provided videos using masked video fine-tuning. *arXiv preprint arXiv:2411.05003*,
1174 2024b.

1175 Haiyu Zhang, Xinyuan Chen, Yaohui Wang, Xihui Liu, Yunhong Wang, and Yu Qiao. 4diffusion:
1176 Multi-view video diffusion model for 4d generation. *arXiv preprint arXiv:2405.20674*, 2024c.

1177 Hao Zhang, Di Chang, Fang Li, Mohammad Soleymani, and Narendra Ahuja. Magicpose4d: Crafting
1178 articulated models with appearance and motion control. *arXiv preprint arXiv:2405.14017*, 2024d.

1180 Kai Zhang, Sai Bi, Hao Tan, Yuanbo Xiangli, Nanxuan Zhao, Kalyan Sunkavalli, and Zexiang Xu.
1181 Gs-irm: Large reconstruction model for 3d gaussian splatting. In *Proc. ECCV*, 2024e.

1182 Richard Zhang, Phillip Isola, Alexei A Efros, Eli Shechtman, and Oliver Wang. The unreasonable
1183 effectiveness of deep features as a perceptual metric. In *Proc. CVPR*, 2018.

1185 Songchun Zhang, Huiyao Xu, Sitong Guo, Zhongwei Xie, Hujun Bao, Weiwei Xu, and Changqing
1186 Zou. Spatialcrafter: Unleashing the imagination of video diffusion models for scene reconstruction
1187 from limited observations. *arXiv preprint arXiv:2505.11992*, 2025a.

1188 Tianyuan Zhang, Hong-Xing Yu, Rundi Wu, Brandon Y Feng, Changxi Zheng, Noah Snavely, Jiajun
1189 Wu, and William T Freeman. Physdreamer: Physics-based interaction with 3d objects via video
1190 generation. In *Proc. ECCV*, 2024f.

1191 Yisu Zhang, Chenjie Cao, Chaohui Yu, and Jianke Zhu. Lion-lora: Rethinking lora fusion to unify
1192 controllable spatial and temporal generation for video diffusion. *arXiv preprint arXiv:2507.05678*,
1193 2025b.

1194 Zhichao Zhang, Hui Chen, Jinsheng Deng, Xiaoqing Yin, Xingshen Song, and Ming Xu. Motion4d:
1195 A decoupled pipeline for enhanced text-to-4d generation with optimized motion patterns. *SSRN*,
1196 2024g.

1197 Yuyang Zhao, Zhiwen Yan, Enze Xie, Lanqing Hong, Zhenguo Li, and Gim Hee Lee. Animate124:
1198 Animating one image to 4d dynamic scene. *arXiv preprint arXiv:2311.14603*, 2023.

1199 Yuyang Zhao, Chung-Ching Lin, Kevin Lin, Zhiwen Yan, Linjie Li, Zhengyuan Yang, Jianfeng
1200 Wang, Gim Hee Lee, and Lijuan Wang. Genxd: Generating any 3d and 4d scenes. *arXiv preprint*
1201 *arXiv:2411.02319*, 2024.

1202 Guangcong Zheng, Teng Li, Rui Jiang, Yehao Lu, Tao Wu, and Xi Li. Cami2v: Camera-controlled
1203 image-to-video diffusion model. *arXiv preprint arXiv:2410.15957*, 2024a.

1204 Yufeng Zheng, Xueting Li, Koki Nagano, Sifei Liu, Otmar Hilliges, and Shalini De Mello. A unified
1205 approach for text-and image-guided 4d scene generation. In *Proc. CVPR*, 2024b.

1206 Junwei Zhou, Xueting Li, Lu Qi, and Ming-Hsuan Yang. Coco4d: Comprehensive and complex 4d
1207 scene generation. *arXiv preprint arXiv:2506.19798*, 2025.

1208 Tinghui Zhou, Richard Tucker, John Flynn, Graham Fyffe, and Noah Snavely. Stereo magnification:
1209 Learning view synthesis using multiplane images. In *SIGGRAPH*, 2018.

1210 Hanxin Zhu, Tianyu He, Anni Tang, Junliang Guo, Zhibo Chen, and Jiang Bian. Compositional
1211 3d-aware video generation with llm director. *arXiv preprint arXiv:2409.00558*, 2024.

1212 Hanxin Zhu, Tianyu He, Xiqian Yu, Junliang Guo, Zhibo Chen, and Jiang Bian. Ar4d: Autoregressive
1213 4d generation from monocular videos. *arXiv preprint arXiv:2501.01722*, 2025.

1214 Chen Ziwen, Hao Tan, Kai Zhang, Sai Bi, Fujun Luan, Yicong Hong, Li Fuxin, and Zexiang Xu.
1215 Long-lrm: Long-sequence large reconstruction model for wide-coverage gaussian splats. *arXiv*
1216 *preprint arXiv:2410.12781*, 2024.

1217
1218
1219
1220
1221
1222
1223
1224
1225
1226
1227
1228
1229
1230
1231
1232
1233
1234
1235
1236
1237
1238
1239
1240
1241

A ADDITIONAL DETAILS OF VIDEO DIFFUSION MODEL

This section elaborates on the GEN3C (Ren et al., 2025) video diffusion backbone introduced in Sec. 3.1 and details how we further strengthen its 3D consistency, on which the fidelity of our distilled reconstruction directly depends.

Conservative mask refinement. A principal limitation of GEN3C’s forward warping is background leakage: occluded object parts in the source view (e.g., the lion body in Fig. 8) become visible from novel viewpoints yet are not flagged in the disocclusion mask, which in turn leaves the foreground incompletely inpainted by the video diffusion model, as shown in Fig. 8 (second and fourth columns).

We address this with a deliberately conservative strategy: because the geometry of occluded regions in $\mathbf{P}^{t,v}$ is unknown, we treat all such regions as occupied and derive refined disocclusion masks accordingly, thereby prompting the video model to reason about potentially visible foreground regions that standard point-based rendering misses.

Concretely, we construct a triangular mesh from the camera-space points $\mathbf{P}_c^{t,v} = \mathbf{C}^t \mathbf{P}^{t,v}$ by connecting spatially adjacent pixels, following Hu et al. (2021):

$$\mathcal{M} = \bigcup_{(u,v)} \{ \triangle(\mathbf{p}_{u,v}, \mathbf{p}_{u+1,v}, \mathbf{p}_{u,v+1}), \triangle(\mathbf{p}_{u+1,v}, \mathbf{p}_{u+1,v+1}, \mathbf{p}_{u,v+1}) \} \quad (2)$$

where $\mathbf{p}_{u,v}$ denotes the 3D point at pixel (u, v) in $\mathbf{P}_c^{t,v}$. This surface mesh acts as a boundary that separates visible from invisible regions. For each rendered pixel we then compare the standard point-based depth $\mathbf{D}^{t,v}$ with the mesh-interpolated depth $\mathbf{D}_{\mathcal{M}}^{t,v}$ obtained by ray-surface intersection:

$$\mathbf{M}^{t,v}(u, v) = \begin{cases} 0 & \text{if } \mathbf{D}_{\mathcal{M}}^{t,v}(u, v) < \mathbf{D}^{t,v}(u, v) - \epsilon \\ \mathbf{M}_{\text{orig}}^{t,v}(u, v) & \text{otherwise} \end{cases}, \quad (3)$$

where $\mathbf{M}_{\text{orig}}^{t,v}$ denotes the original disocclusion mask from forward warping and ϵ is a small tolerance. Whenever the mesh surface lies closer than the original points, we conservatively mask the region as a potential foreground disocclusion. This yields markedly more reliable guidance for video generation, particularly under large viewpoint changes, as shown in Fig. 8 (third and fifth columns), and correspondingly improves the consistency of the supervision our decoder distills from.

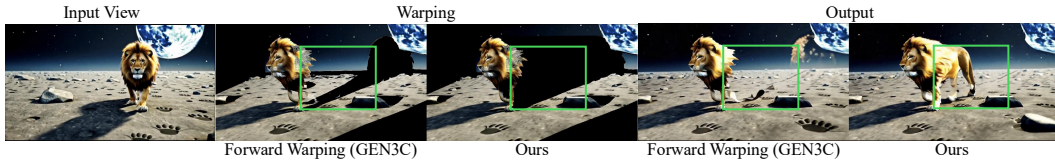


Figure 8: **Rendering function comparison.** Our model builds upon GEN3C (Ren et al., 2025), which conditions camera control on forward-warped images. We contrast forward warping (left) with our conservative rendering function (right): the highlighted region is erroneously filled with background pixels under forward warping but correctly masked by ours, so that the video model completes the disoccluded foreground faithfully rather than leaking background.

B MODEL DETAILS

Progressive training. Training the 3DGS decoder directly on 6 trajectories of 121 frames at 704×1280 is prohibitively expensive, so we adopt a progressive training schedule, detailed in Tab. 3, that increases spatial resolution, temporal length, and view count in stages, and we initialize the dynamic model from the final static pre-trained model. Total training takes 6 days on 8 NVIDIA A100 80 GB GPUs, with gsplat (Ye et al., 2025) as the 3DGS implementation.

Subsampling Gaussians. The count of pixel-aligned 3D Gaussians scales linearly with both spatial and temporal resolution: our generated scenes span 726 frames at 704×1280 , which would entail 654,213,120 per-pixel Gaussians. We therefore emit a single 3D Gaussian per 8×8 spatial neighborhood, reducing the total by a factor of 64 to 10,222,080, and our opacity-based pruning

Table 3: Progressive training setup. Training proceeds in stages that progressively increase the spatial resolution $H \times W$, the temporal length L , and the number of views V , before culminating in the dynamic stage that fine-tunes from the final static model. This schedule makes the 6×121 -frame, 704×1280 regime tractable within a fixed compute budget.

Stage	$H \times W$	L	V	S	B	Steps
Static						
1	176×320	17	1	17	4	10k
2	176×320	49	1	49	4	2.5k
3	352×640	49	1	49	2	2.5k
4	704×1280	49	1	49	1	2.5k
5	704×1280	121	1	9	1	57.5k
6	704×1280	121	1–6	9	1	7k
Dynamic						
7	704×1280	121	6	12	1	10k

contracts this further to 2,044,416—a $320\times$ reduction overall that yields a substantially more compact representation while preserving rendering quality.

Joint Transformer-Mamba blocks. We adopt the reconstruction block of Long-LRM (Ziwen et al., 2024): one Transformer layer followed by seven Mamba-2 layers, stacked twice to give 16 layers with a 512-dimensional hidden state. Our network is thus considerably smaller than those of Long-LRM (Ziwen et al., 2024) and Wonderland (Liang et al., 2025a), which use 24 layers at 1024 hidden dimensions, yet it attains the reported quality at a fraction of the parameter count. While pure Transformers are generally more expressive, they are markedly slower in both training and inference than Mamba-2. Since we trained both configurations under an identical budget, we attribute the residual gap to optimization and infer that Transformer-only blocks would require substantially more iterations to match the rendering quality of the joint blocks.

Camera encoding. Following prior work (Kant et al., 2024; Zhang et al., 2024e), we encode cameras with Plücker embeddings. The embeddings are computed at full spatial and temporal resolution and projected into the video latent space by the pre-trained RGB encoder, so that camera geometry and appearance share a common latent representation. Because that encoder expects 3 input channels while the Plücker embeddings are 6-dimensional, we encode the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins separately, then concatenate the two latent encodings along the channel dimension.

Time encoding. Analogously, to render the 1-dimensional time input compatible with the encoder’s 3-channel interface, we concatenate a 2-dimensional sinusoidal embedding to the raw time value along the channel dimension, replicate the result across the spatial dimensions, and encode it into the video latent space. This procedure encodes the source and target times independently within the compressed latent representation, exactly mirroring the camera encoding above.

Dynamic data augmentation. Fig. 9 visualizes the dynamic data augmentation procedure for two example camera trajectories. The procedure constructs pairs of supervision views that observe the scene from the same motion state but from opposite extreme viewpoints, thereby guaranteeing balanced spatial coverage at every timestep.

C ADDITIONAL EVALUATION

C.1 ADDITIONAL BASELINE DETAILS

We additionally compare against BTimer (Liang et al., 2025b), a recent method for joint 3D and 4D reconstruction from posed images or videos. To ensure a fair comparison, we asked the BTimer authors to run experiments on the same scenes as *Lyra*. Since BTimer is purely regression-based, we pair it with our GEN3C (Ren et al., 2025) video diffusion backbone: we use identical camera trajectories and feed the RGB-decoded GEN3C videos as input to BTimer, so that both methods reconstruct from the same generated content and differ only in the reconstruction stage. BTimer operates in pixel space and expects 12 input images; as noted in the main paper, however, pixel-space models such as BTimer exhaust memory when handed the full 726 high-resolution GEN3C frames

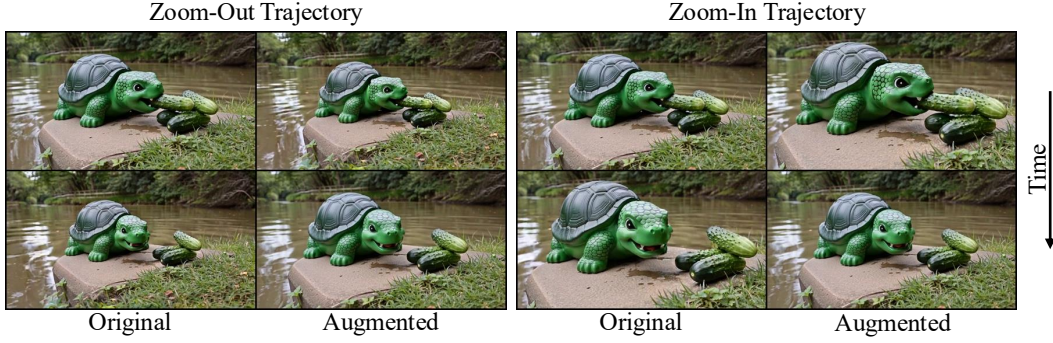


Figure 9: **Dynamic data augmentation videos.** We augment the supervision with a motion-reversed video so that every timestep is observed with full spatial coverage, thereby forestalling the low-opacity artifacts that otherwise arise at early timesteps. For two example trajectories—zoom-out and zoom-in—we visualize the corresponding augmented videos, whose camera motion is reversed relative to the originals.



Figure 10: **Image-to-3DGS Comparison.** We compare our method with BTimer (GEN3C) under an identical GEN3C backbone, showing five views rendered from the generated scenes. Our results exhibit substantially fewer artifacts and markedly higher fidelity.

directly. We therefore uniformly subsample the GEN3C frames: for static evaluations we select 12 frames that evenly span the available viewpoints, and for dynamic evaluations we sample 11 frames across the viewpoint range together with the bullet-time frame of the motion being reconstructed, guaranteeing that the target motion state is always present as a reference. We denote this integrated baseline BTimer (GEN3C), i.e., BTimer combined with GEN3C.

C.2 ADDITIONAL STATIC 3D EVALUATION

For image-to-3DGS generation, Fig. 10 presents qualitative comparisons with BTimer (GEN3C) and Tab. 4a the corresponding quantitative results. Under an identical GEN3C backbone, our method attains substantially higher fidelity with markedly fewer artifacts, improving PSNR from 16.32 to 24.92—an 8.60 dB gain that isolates the benefit of our latent-space, self-distilled decoder over a pixel-space regression baseline.

Table 4: **Quantitative results on the static and dynamic *Lyra* datasets.** Across both settings, and holding the GEN3C backbone fixed for both methods, our method consistently surpasses BTimer (GEN3C) in PSNR, SSIM, and LPIPS; the best value in each column is shown in **bold**.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
BTimer (GEN3C)	16.32	0.580	0.427	BTimer (GEN3C)	20.29	0.687	0.315
Ours	24.92	0.834	0.183	Ours	23.07	0.779	0.231

(a) **Static *Lyra* dataset.** Under an identical GEN3C backbone, our method outperforms BTimer (GEN3C) on all three metrics.

(b) **Dynamic *Lyra* dataset.** On dynamic scenes, our method again outperforms BTimer (GEN3C) on all three metrics.

C.3 DYNAMIC 3D EVALUATION

BTimer (GEN3C) comparison. For dynamic 3D evaluation, we compare against BTimer (GEN3C) (Sec. C.1) on 100 out-of-distribution videos from our dynamic *Lyra* dataset, cropping each video to 512×512 for a fair comparison, since BTimer was trained predominantly at that resolution. As reported in Tab. 4b, our method substantially outperforms BTimer (GEN3C), raising PSNR from 20.29 to 23.07 and confirming that the advantage observed in the static setting carries over to dynamic scenes.

C.4 DEPTH VISUALIZATION

Fig. 11 visualizes depths recovered from our generated 3DGS. Depth supervision prevents flattened geometry while leaving visual quality entirely intact.

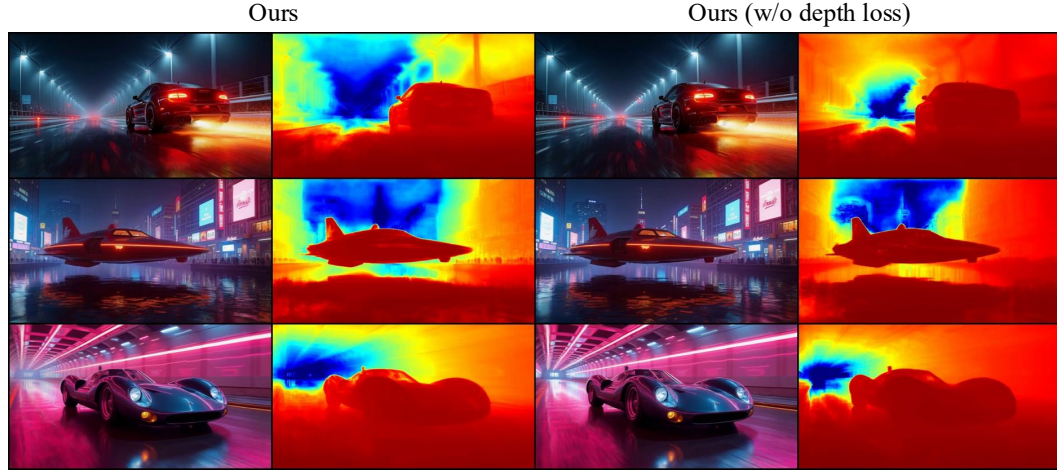


Figure 11: **Depth loss ablation.** 3DGS renderings and their corresponding depth maps. Adding the depth loss as auxiliary supervision prevents flattened geometry while leaving visual quality unaffected, so geometric fidelity is gained at no perceptual cost.

D ROBOT SIMULATION

A primary motivation for building a diverse 3D generator is its capacity to synthesize simulation environments for autonomous agents. To this end, we develop an end-to-end pipeline: we generate 3D scenes from text with our model, export the 3D Gaussians as a .ply file, and convert it to .usdz format using 3DGUT (Wu et al., 2025a) and its export function. The resulting .usdz file imports directly into the NVIDIA Isaac robot development platform as a physically based virtual environment, within which AI-based robots can be trained and tested under diverse conditions. This pipeline demonstrates that our generated scenes are not merely renderable but are directly usable as physically grounded simulation assets. Early demo results will be released upon acceptance; a screenshot of the interface is shown in Fig. 12.

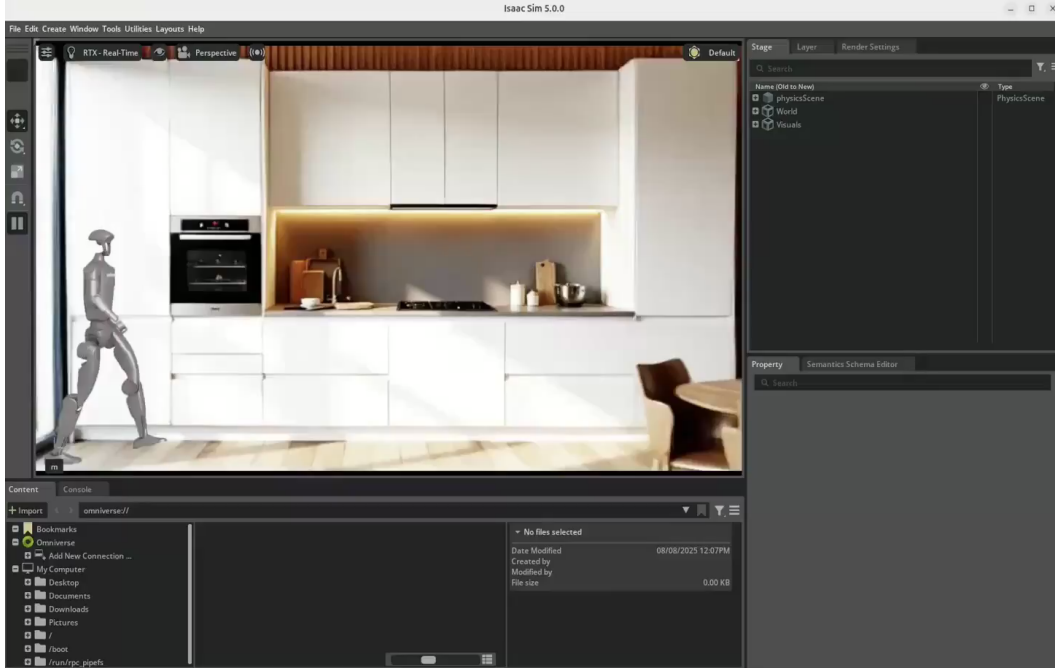


Figure 12: **Robot Simulation.** A frame of a robot simulation running in the Isaac Sim 5.0 framework, using a 3DGS scene generated by our method as the input environment, illustrating that our outputs serve directly as physically grounded simulation assets.

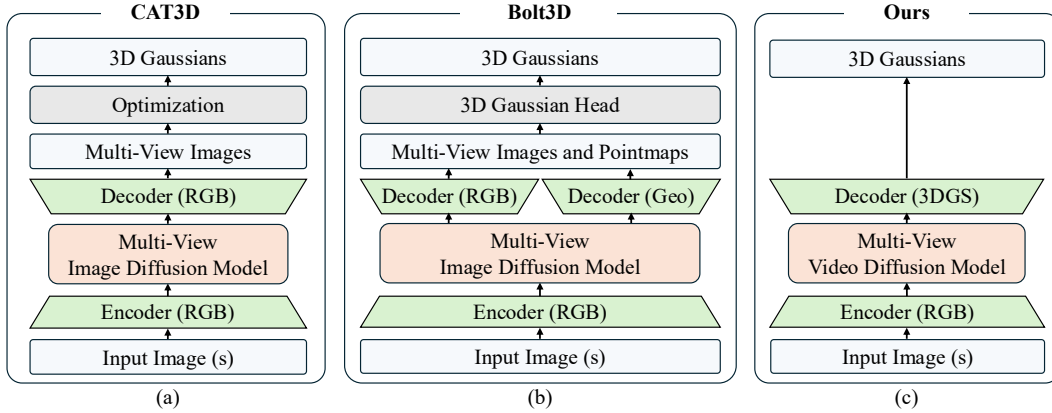


Figure 13: **Approaches for 3D Generation.** CAT3D (Gao et al., 2024b) proposes a multi-view image diffusion model that renders novel viewpoints, which a subsequent optimization stage reconstructs into 3D Gaussians. Bolt3D (Szymanowicz et al., 2025b) fine-tunes CAT3D to emit pointmaps through a geometry autoencoder and, in place of optimization, predicts the 3D scene with a feed-forward 3D Gaussian head. Our work, by contrast, builds upon a pre-trained camera-controlled video diffusion model and *directly* decodes the multi-view video latents into 3D Gaussians, dispensing with both the separate geometry autoencoder and the multi-view image intermediary.

E RELATED WORK

This section surveys additional related work and, in Fig. 13, contrasts our approach with the prior methods CAT3D (Gao et al., 2024b) and Bolt3D (Szymanowicz et al., 2025b), situating our contribution within the broader landscape.

3D generation. Early 3D generation centered on single object categories, extending GANs into 3D by employing neural renderers as an inductive bias (DeVries et al., 2021; Chan et al., 2022; Or-EI et al., 2022; Schwarz et al., 2022; Bahmani et al., 2023b). The advent of CLIP-based supervision (Radford

et al., 2021) shifted the field toward more flexible and diverse asset creation, enabling both text-driven generation and editing (Chen et al., 2018; Jain et al., 2022; Sanghi et al., 2022; Jetchev, 2021; Gao et al., 2023; Wang et al., 2022). More recently, diffusion models have sharply improved quality by supplanting CLIP guidance with Score Distillation Sampling (SDS) (Poole et al., 2023; Wang et al., 2023b; Lin et al., 2023a; Chen et al., 2023b; Liang et al., 2023; Wang et al., 2023a; Li et al., 2024e; He et al., 2024b; Ye et al., 2024; Liu et al., 2024b; Yu et al., 2023; Katzir et al., 2024; Lee et al., 2024a; Sun et al., 2024a).

To reinforce structural coherence, several methods enforce multi-view consistency by generating scenes from distinct perspectives (Lin et al., 2023b; Liu et al., 2023; Shi et al., 2024; Feng et al., 2024a; Liu et al., 2024a; Kim et al., 2023; Voleti et al., 2024; Höllein et al., 2024; Tang et al., 2024c; Gao et al., 2024b; Wang et al., 2025f; Kant et al., 2025; Yuan et al., 2025; Ren et al., 2024b), whereas others cast scene synthesis as iterative inpainting (Höllein et al., 2023; Shriram et al., 2024). A further line of work lifts 2D images into 3D using NeRF (Mildenhall et al., 2020), 3D Gaussian Splatting (Kerbl et al., 2023), or mesh-based representations coupled with diffusion models (Chan et al., 2023; Tang et al., 2023a; Gu et al., 2023; Liu et al., 2024c; Yoo et al., 2023; Tewari et al., 2023; Qian et al., 2024b; Long et al., 2024; Wan et al., 2024; Szymanowicz et al., 2023; Lu et al., 2024).

Feed-forward 3D models. A complementary body of work develops fast, feed-forward techniques that predict 3D content directly from images or text (Hong et al., 2024; Li et al., 2024b; Xu et al., 2024e; Zhang et al., 2024a; Han et al., 2024; Jiang & Wang, 2024; Xie et al., 2024a; Tang et al., 2024b; Tochilkin et al., 2024; Qian et al., 2024a; Szymanowicz et al., 2024; 2025a; Liang et al., 2025a; Szymanowicz et al., 2025b; Schwarz et al., 2025; Yang et al., 2025; Zhang et al., 2025a). Unlike our method, however, these approaches remain confined to static 3D scenes, and some target narrow domains such as faces (Kirschstein et al., 2025). Concurrent work (Liang et al., 2025b; Xu et al., 2025) addresses real scenes but cannot accommodate the diverse generated scenes or large viewpoint changes that our framework handles.

4D generation. Progress in 4D generation, i.e., dynamic 3D scene synthesis, has been rapid, with many approaches guided by input text prompts or images. Following the introduction of large-scale generative models for the task (Singer et al., 2023), subsequent work has advanced both visual fidelity and motion quality (Ren et al., 2023; Ling et al., 2024a; Bahmani et al., 2024b; Zheng et al., 2024b; Gao et al., 2024a; Yang et al., 2024a; Jiang et al., 2024b; Miao et al., 2024; Li et al., 2024a; Zhang et al., 2024g; Yuan et al., 2024; Jiang et al., 2024a).

Beyond text conditioning, an alternative family of methods lifts 2D images or videos into dynamic 3D scenes (Ren et al., 2023; Zhao et al., 2023; Yin et al., 2023; Pan et al., 2024; Zheng et al., 2024b; Ling et al., 2024a; Gao et al., 2024a; Zeng et al., 2024b; Chu et al., 2024; Wu et al., 2024; Yang et al., 2024b; Wang et al., 2024c; Feng et al., 2024b; Sun et al., 2024c; Zhang et al., 2024c; Ren et al., 2024a; Lee et al., 2024b; Li et al., 2024c; Van Hoorick et al., 2024; Uzolas et al., 2024; Chai et al., 2024; Liang et al., 2024; Li et al., 2024d; Xie et al., 2024b; Li et al., 2025a; Wang et al., 2025e; Zhu et al., 2025; Zhou et al., 2025; Hu et al., 2025; Park et al., 2025). Further efforts inject physics priors into the generation pipeline (Lin et al., 2024; Huang et al., 2024b; Zhang et al., 2024f; Kiray et al., 2025) or sharpen motion controllability through template-driven approaches (Zhang et al., 2024d; Sun et al., 2024b; Chen et al., 2024b), while a parallel direction pursues compositional and interactive 4D generation (Bahmani et al., 2024a; Xu et al., 2024b; Cao et al., 2024; Yu et al., 2024a; Zeng et al., 2024a; Zhu et al., 2024).

A separate strand extends 3D GAN frameworks into the 4D setting by training directly on 2D video (Bahmani et al., 2023a; Xu et al., 2023). Such methods, however, are typically constrained by small-scale, single-category datasets that limit generalization, and most remain object-centric, neglecting background. Their visual fidelity consequently trails the photorealism of recent video generation models—precisely the source of supervision our method exploits, and the reason our framework generalizes to diverse scenes where these approaches do not.

Camera-conditioned video models. A substantial body of recent work incorporates camera control into video diffusion models. The pioneering MotionCtrl (Wang et al., 2024e) introduced camera conditioning by augmenting pre-trained video models (Chen et al., 2023a; Blattmann et al., 2023) with extrinsic matrices. Subsequent efforts (He et al., 2024a; Xu et al., 2024c; Kuang et al., 2024; Ju et al., 2025; He et al., 2025; Li et al., 2025c; Wang et al., 2025c; 2024d; Lei et al., 2025; Liu et al., 2025) strengthened this conditioning through Plücker-coordinate camera representations, while

1566 others follow MotionCtrl in simply flattening the poses (Wang et al., 2025b). A further line (Hu et al.,
1567 2024; Xiao et al., 2024; Ling et al., 2024c; Hou et al., 2024) achieves camera-motion control without
1568 introducing additional trainable parameters, and yet others adopt low-rank formulations for camera
1569 control (Zhang et al., 2025b). All of these, however, rest on U-Net-based architectures and, critically,
1570 leave their outputs ungrounded in 3D.

1571 Several recent methods continue to extend U-Net-based camera control (Zhao et al., 2024; Xu et al.,
1572 2024a; Zheng et al., 2024a; Zhang et al., 2024b; Yu et al., 2024b; Lei et al., 2024), while others begin
1573 to explore diffusion transformers. Cheong et al. (Cheong et al., 2024) propose one such transformer-
1574 based framework, though its scene and visual quality remain limited, and DimensionX (Sun et al.,
1575 2024d) introduces joint space-time control in diffusion transformers but is constrained to pre-defined,
1576 non-continuous camera trajectories. Others instead pursue pose and geometry estimation with a video
1577 DiT (Huang et al., 2024a; Mai et al., 2025; Jiang et al., 2025b). CAT4D (Wu et al., 2025b) derives a
1578 multi-view video diffusion model by fine-tuning a multi-view backbone, and large-scale synthetic
1579 data further boosts performance, as SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al.,
1580 2025a) demonstrate with transformer-based architectures. Finally, 4DiM (Watson et al., 2024) trains
1581 a space-time diffusion model from scratch for single-image novel-view synthesis. A related direction
1582 edits the camera movement of videos with underlying motion via camera retargeting (Seo et al., 2025;
1583 Ma et al., 2025), and a particularly influential one conditions camera-controlled video generation on
1584 a spatiotemporal cache built from point clouds or optical flow (Wu et al., 2025c; Ren et al., 2025;
1585 Wang et al., 2025d; Cao et al., 2025; YU et al., 2025; Jin et al., 2025; Xing et al., 2025; Yan et al.,
1586 2025; Li et al., 2025b; Gu et al., 2025).

1587
1588
1589
1590
1591
1592
1593
1594
1595
1596
1597
1598
1599
1600
1601
1602
1603
1604
1605
1606
1607
1608
1609
1610
1611
1612
1613
1614
1615
1616
1617
1618
1619